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Numerical Spectral Correspondence Between a Non-Autonomous Quadratic Map and the Riemann Zeros: An Exploratory Study

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Abstract

This study examines a non-autonomous quadratic map driven by a logarithmic cooling schedule ($\mu_n \sim 1/\ln^2 n$, a phenomenological ansatz), building on our recent published result that the logistic map's symbolic dynamics at its band-merging point is isomorphic to the prime sieve. From its trajectories, we construct an empirical, non-normal, dissipative transfer matrix and compares its complex eigenphases to the non-trivial Riemann zeros after calibrating a few free parameters against the same low-order zeros—an in-sample numerical correspondence, not an independent prediction. We quantify this gap directly: fitting on the first $M \in \{50, 70, 80\}$ zeros and evaluating on the rest gives a held-out MSE one to two orders of magnitude larger than the training error, with the fitted coupling drifting across M but remaining comparatively stable across ten random seeds at fixed $M = 70$ (CV $\approx 4.7\%$). At low order ($N \lesssim 20$), an unselected re-computation shows a qualitative rank correlation ($\rho = 0.56$, $p = 0.010$) with, but no significant joint co-location ($p = 0.099$) of, a residual feature reported in recent ion-trap quantum simulations of the same zeros. At larger N ($N \geq 1000$), the model matches a globally rescaled GUE surrogate's mean counting-function trend better once conjugate eigenphases are restored, though this partly follows from the construction's own symmetry; a separate, standard unfolded local-statistics test shows the model's own eigenphase spacings do *not* match GUE, unlike the true zeros. These are numerical observations on a heuristic model, not a proof or Hilbert–Pólya-type operator construction; a dedicated table tabulates the epistemic status of every main claim.

Keywords: Riemann zeros; non-autonomous dynamical systems; logistic map; transfer operator; empirical spectral correspondence; numerical continuation

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1. Introduction

Since the seminal hypothesis by Riemann [1], finding a Hermitian operator whose eigenvalues correspond to the imaginary parts of the non-trivial Riemann zeros has always been the "central open problem" in the field of mathematical physics (i.e., the famous Hilbert-Pólya conjecture; see [2–4] for reviews), and the search for such a concrete operator remains active, e.g., the recent explicit Hamiltonian construction of Yakaboylu [5]. The zeros themselves are inseparable from the classical theory of prime counting: their existence and analytic continuation underlie the Prime Number Theorem, independently

established by Hadamard [6] and de la Vallée Poussin [7], and their finer statistics touch on the Hardy–Littlewood conjectures for prime constellations [8]. The classical analytic theory of $\zeta(s)$ itself is treated comprehensively in Titchmarsh's monograph [9], and large-scale numerical verification of the zeros' locations was pioneered by Odlyzko [10], with such high-precision numerical evaluation of $\zeta(s)$ still an active line of methodological work today [11]. Over half a century after Riemann, the pioneering work of Montgomery [12] and numerical verifications by Odlyzko [13] established a notable connection between the Riemann zero spacing and the Gaussian Unitary Ensemble (GUE) of Random Matrix Theory (RMT), whose statistical machinery traces back to Wigner's semicircle law for nuclear energy levels [14] and was later codified in Mehta's treatise [15]. This spacing statistics connection was subsequently generalized by Katz and Sarnak to families of L-functions [2,16] and refined through the moments conjecture of Keating and Snaith [17], connecting this number-theoretic problem to the physics of Quantum Chaos [18]. On the physics side, the semiclassical link between chaotic spectra and RMT statistics is the content of the Bohigas–Giannoni–Schmit conjecture [19], itself grounded in Gutzwiller's periodic-orbit trace formula [20] and its number-theoretic analogue, the Selberg trace formula [21]; by contrast, generic integrable systems display Poissonian level statistics [22]. Subsequently, Sir Michael Berry and Keating proposed the renowned H = xp quantum Hamiltonian conjecture [23], later developed by Sierra and collaborators into explicit spectral realizations of the Riemann zeros [24,25]. Other models, such as those based on non-commutative geometry [26], have also been explored. However, existing analytical models generally derive the macroscopic statistical laws (GUE distribution) of high-order zeros well, but rarely generate the specific, discrete coordinates of low-order zeros deterministically from first principles. Constructing a deterministic dynamical system that reproduces the true Riemann spectrum from first principles remains an open problem.

In recent years, with advances in quantum information technology, physicists have begun attempting to "physically" find Riemann zeros in the laboratory [27]. For example, a team from the University of Science and Technology of China (USTC) measured the first 80 Riemann zeros on physical equipment using Floquet engineering by periodically driving qubits in an Ion Trap with microwaves [28]. Although the low-frequency band achieved high fidelity, the experiment simultaneously reported an unexplained feature: when approaching specific energy levels (e.g., $N \approx 20$), the measurement deviation of the physical instruments showed a large divergence (enlarged error bars) that departed from a conventional Gaussian background. Floquet-engineered simulation of driven quantum systems has itself continued to scale on near-term hardware since then [29], underscoring that the platform is an active target for larger-scale zero-finding experiments rather than a one-off demonstration. The physics community still lacks a unified theoretical framework to explain this: whether it is due purely to instrumental engineering decoherence such as laser linewidth, or in part to some intrinsic limit encountered by the system when approximating highly nonlinear manifolds remains an open question we do not resolve here—a question that touches upon the fundamental asymptotics of eigenvalue distribution originally discussed by Weyl [30].

Rather than starting from an analytically motivated Hamiltonian, we take an exploratory, "bottom-up" route, motivated by our recent finding that the symbolic dynamics of the Logistic map at its band-merging point is isomorphic to the prime sieve, with the Hardy–Littlewood twin-prime constant emerging as a fixed point of the dynamics [31]. Since that work connects a deterministic map to the arithmetic of the primes, it is natural to ask, purely as a numerical experiment, what the eigenphase spectrum of an empirical transfer operator built from a related map looks like, and whether it bears any quantitative resemblance to the Riemann zeros; we emphasize at the outset that this question, and ev-

erything that follows from it, is explored numerically and heuristically, and that we do not construct or claim to approximate a self-adjoint Hilbert–Pólya operator. The approach taken here also draws on the classical observations that simple maps can have complicated dynamics [32,33] and on the universality of nonlinear transformations [34], independently derived via renormalization-group iteration by Coullet and Tresser [35] and systematized in the reprint collection of Cvitanović [36]; the underlying combinatorial structure of the period-doubling cascade itself was first classified by Metropolis, Stein and Stein [37]. Concretely, we discard the autonomous static potential and constructs a discrete Markov evolution operator with non-autonomous (time-dependent) parameter, in the spirit of Ulam and von Neumann's original proposal to approximate deterministic chaotic maps by finite Markov chains [38,39], whose ergodic-theoretic justification (existence of absolutely continuous invariant measures for piecewise-monotonic maps) is due to Lasota and Yorke [40]; broader treatments of quantum chaos and its RMT signatures appear in [41,42]. Using large-scale parallel computation and Arnoldi iteration [43], we search at high resolution ($\epsilon \approx 10^{-3}$) for dynamical trajectories that trigger a sharp numerical convergence transition in the system.

Next, we present the non-autonomous dynamical model. We report numerical correspondences between an empirical transfer-matrix spectrum and the Riemann zeros across scales, from low-order eigenphase alignment (obtained after calibrating a small number of free parameters against the same low-order zeros) to macroscopic asymptotic behavior. We then examine the model's dependence on numerical resolution and discusses how breaking the intrinsic conjugate symmetry of the eigenphase spectrum (a consequence of P being a real-valued matrix) affects the macroscopic fit. A dedicated table in Section 6 states the epistemic status of each claim explicitly.

2. The Mathematical Model

2.1. The Dynamic Kernel: Non-Autonomous Logistic Map

The present work considers a classical, deterministic non-autonomous dynamical system on the interval $[-1, 1]$; no Hilbert-space operator or unitary evolution is postulated or constructed at any point in this work. The microscopic evolution is governed by the canonical Non-autonomous Quadratic Map (topologically conjugate to the Logistic Map):

$$x_{n+1} = 1 - \lambda_n x_n^2. \tag{1}$$

In standard chaos theory, the control parameter λ is typically a constant (e.g., the Feigenbaum accumulation point $\lambda_\infty \approx 1.401$). However, to match the logarithmic density distribution of Riemann zeros (Weyl's Law), we must break the system's time-translation symmetry. We introduce an explicit non-autonomous drive mechanism where the control parameter λ_n evolves with the iteration step (energy scale) n:

$$\lambda_n = \mu_c + \delta_n. \tag{2}$$

Here, $\mu_c \approx 1.5437$ is a fixed baseline value of the control parameter (the "critical baseline"). Unlike the standard edge-of-chaos threshold, this value is empirically anchored by the energy of the first non-trivial zero ($E_1 \approx 14.13$), ensuring the spectral accuracy of the low-lying modes.

2.2. A Logarithmic Cooling Ansatz and Higher-Order Corrections

The core hypothesis, presented here as a phenomenological ansatz and not as a derived result, is that the microscopic perturbation term δ_n should follow a decay law combining the known asymptotic Riemann-zero spacing $\Delta E \propto 1/\ln n$ (Weyl's Law) with an assumed

square-root relation between a parameter perturbation and the resulting eigenphase spacing. Section 3.1 gives the full heuristic chain leading to the leading-order form $1/\ln^2 n$ used below; we do not derive it from a theorem about the eigenphase spacing of the discretized transfer matrix, and we do not claim that alternative logarithmic forms are excluded.

However, extending this macroscopic evolution into large-N regimes necessitates accounting for intense nonlinear transient fluctuations near the first non-trivial zero. Therefore, rather than relying on localized empirical fits, we expand the macroscopic coupling strength with an additional empirical correction term. The Unified Dynamic Equation of the system is expressed as follows:

$$x_{n+1} = 1 - \underbrace{\left(\mu_c + \frac{k_1}{\ln^2 n} + \frac{k_2}{\ln^3 n}\right)}_{\lambda_n^{\text{eff}}} x_n^2. \tag{3}$$

Here λ_n^{eff} denotes the *effective* (iteration-dependent) control parameter actually applied at step n: the constant baseline μ_c plus the two running corrections. Equation (3) thus makes explicit that (2) is realized with $\delta_n = k_1/\ln^2 n + k_2/\ln^3 n$.

This explicit construction motivates a tripartite decomposition of the fitted parameter schedule:

Baseline Term (μ_c): This constant term fixes the system's operating point near the critical threshold.

Leading Correction ($k_1/\ln^2 n$): The dominant logarithmic decay term makes the control parameter depend explicitly on the iteration step n, driving the system to approach the critical attractor at the rate required to match the macroscopic growth of the Riemann zero counting function.

Higher-Order Correction ($k_2/\ln^3 n$): This second-order term is included empirically to absorb curvature distortions that appear early in the cooling schedule, so that the fitted value of the leading term k_1 is less sensitive to those early-time transients.

We emphasize that this is an empirical fit, not a result extrapolated from fixed-point theory or any other theorem: the values of k_1 and k_2 are determined by minimizing a loss against the Riemann zeros themselves, using the Differential Evolution (DE) algorithm detailed in the Parameter Optimization Methods (Appendix A). Because the same zeros used to fit these parameters are also used to assess the resulting agreement, this fit does not by itself constitute independent evidence of predictive power; we return to this limitation in the Discussion (Section 5).

3. Methods

3.1. Dynamical Core Selection, Scaling Law, and Topological Constraints

The Choice of the Dynamical Core

In exploring possible dynamical analogues for the microscopic structure of the Riemann zeros, this study avoids complex, analytically motivated quantum Hamiltonians. Instead, we select the most fundamental nonlinear quadratic map—the Logistic map ($x_{n+1} = 1-\mu x_n^2$)—as the computational core. We use our empirically anchored baseline ($\mu_c \approx 1.543689$; distinct from the Feigenbaum accumulation point) as the reference point about which the non-autonomous parameter evolves. This is an exploratory numerical construction, not a mapping between the Riemann zeros and a self-adjoint Hilbert-Pólya operator.

Motivation from the Prior Prime-Sieve Statistical Correspondence

The choice of this specific dynamical core, and specifically of $\mu_c \approx 1.543689$ as its reference point, is not arbitrary: it is inherited directly from our prior study [31], which is heuristic and numerical in nature but establishes two distinct kinds of evidence that make u_c a distinguished parameter value rather than a free choice. The first is a topological (symbolic-dynamics) admissibility argument, summarized precisely in Appendix A. The second, and the one most directly relevant to the present paper's methodology, is a purely *statistical* correspondence obtained by treating the prime sequence and the autonomous Logistic orbit at u_c as two symbolic time series and comparing their empirical statistics side by side, with no shared parameters and no fitting between them: (i) the discrete gap-density spectrum of consecutive prime gaps and of consecutive "surviving" symbols in the chaotic orbit both show pronounced resonance peaks at multiples of 6, with a Pearson correlation coefficient above 0.99 between the two discrete spectra; (ii) both sequences show a positive but strongly suppressed maximal Lyapunov exponent relative to fully developed chaos, i.e., a "weak chaos" or "edge of chaos" signature, rather than the signature of either a regular or a purely random process and (iii) their block-entropy growth and asymptotic entropy rate are of the same order ($\approx$0.6 bits per symbol), indicating a comparable information-theoretic complexity, or "grammar," of the two sequences. None of these three comparisons is a proof that the two sequences are identical, conjugate, or isomorphic in any precise sense; they are numerical observations of statistical similarity, and we use that weaker, more accurate description of them here in place of the stronger topological language used elsewhere.

Taken together, this prior statistical correspondence is the reason the present paper singles out u_c, rather than some other parameter value or some other one-dimensional map, as the dynamical core to carry forward. The logic is a heuristic bridge, not a derivation: if one specific chaotic attractor's own orbit statistics already resemble the statistical texture of the primes, and the classical analytic theory of $\zeta(s)$ ties the distribution of primes to the distribution of the non-trivial zeros through the explicit formula, then it is a natural—but not logically forced—next question to ask whether the *spectral* structure built from that same dynamical core (rather than its raw orbit statistics) shows any numerical correspondence with the zeros themselves. This is the question the remainder of the paper investigates. We are explicit that this bridge argument is only a motivation for the construction, and that the numerical correspondences reported in Sections 4–6 must be judged on their own evidence (and limitations), independently of whether the motivating heuristic itself is eventually made rigorous.

Heuristic Motivation for the Scaling Law

A static autonomous system (with a fixed μ) can only generate a static snapshot of spectral lines. To reproduce an infinitely extending spectrum with continuously increasing density, the system must be non-autonomous, cooling dynamically toward U_c as the energy level n progresses. The following chain of heuristic correspondences motivates, but does not derive from first principles, a particular form for the parameter perturbation $\Delta\mu$:

The Riemann Target: The Riemann–von Mangoldt formula dictates that the local average spacing ΔE between adjacent zeros decays logarithmically: $\Delta E \propto 1/\ln n$.

The Dynamical Constraint: Due to the parabolic extremum of the quadratic map at $x = 0$, classical scaling laws impose a square-root relationship between the parameter perturbation $\Delta\mu$ and the eigenphase spacing $\Delta\Phi$ of its transition matrix: $\Delta\Phi \propto \sqrt{\Delta\mu}$.

A Heuristic Correspondence: As a proposed analogy, to obtain a close topological correspondence we impose $\Delta\Phi \propto \Delta E$. Combining the constraints yields $\sqrt{\Delta\mu} \propto 1/\ln n$, which squares to the adiabatic cooling form:

$$\Delta\,\mu \propto \frac{1}{\ln^2 n} \tag{4}$$

Within this heuristic chain, alternative forms like $1/\ln n$ would, after the same square-root step, compress the spectrum too weakly ($1/\sqrt{\ln n}$), producing larger errors in the large-N region under the same fitting protocol. We do not claim this constitutes a proof that such alternative forms are excluded on independent theoretical grounds, only that $1/\ln^2 n$ performs better empirically among the forms we tried.

Choice of Direction of Evolution

Beyond the $1/\ln^2 n$ cooling rate motivated above, the *direction* of approach toward U_c is a further modeling choice, not a derived necessity. We adopt the control equation $\mu_n = U_c + k(n)/\ln^2(n+c)$, which keeps $\mu_n > U_c$ at all times.

Approaching from the left ($\mu < U_c$) would trap the system in the stable period-doubling cascade regime. Here, the system exhibits regular predictability, and the eigen-spectrum of its transfer matrix degenerates into a small number of discrete, simple points with low spectral complexity. Conversely, the right side ($\mu > U_c$) is the chaotic band-merging regime, where the empirical transfer operator generates a much richer complex eigenphase spectrum as the parameter is annealed toward the edge of chaos. We approach from this side because it is the regime in which the eigenphase spectrum is rich enough, numerically, to be fitted against the Riemann zeros; we do not claim this choice is uniquely dictated by any established mathematical necessity, only that it is the empirically productive direction for the fits reported below.

3.2. Module A: Spatial Discretization Operator

To extract a discrete complex eigenvalue spectrum from continuous non-autonomous dynamics, we must map the infinite-dimensional continuous phase space to a finite-dimensional discrete transfer matrix (Frobenius-Perron Operator). Of the four discretization schemes we compared (a raw Monte Carlo trajectory count, a naïve point-mapped Ulam scheme, the exact geometric-overlap Ulam method, and a Gaussian-kernel splatter; full comparison in Appendix B.2), the Gaussian-kernel splatter is used throughout this study: after the particle transitions to the target area, it is no longer an absolute singularity but spreads into a Gaussian distribution, smoothly diffusing into the target neighborhood. This combines the efficiency of point mapping with the need for operator smoothing, accelerating matrix assembly while damping fine-scale discretization artifacts. After the phase space is discretized into N grids with center coordinates c_i, the Gaussian-broadened transition probability P_{ij} for a particle transitioning from source grid i to target grid j is defined as follows:

$$P_{ij} = \frac{1}{Z_i}\exp\left(-\frac{(c_j - f(c_i))^2}{2\sigma^2}\right), \tag{5}$$

where the partition function $Z_i = \sum_{k=1}^{N} \exp\left(-\frac{(c_k - f(c_i))^2}{2\sigma^2}\right)$ ensures local probability conservation.

3.3. Module B: Temporal Evolution Integration

The core challenge of non-autonomous systems is that the transfer matrix P_t is different at every moment. Two more direct approaches—accumulating the raw product $\prod_t P_t$,

and tracking the evolution via continuous QR decomposition—both fail for the reasons detailed in Appendix B.3 (floating-point underflow, and loss of accumulated phase information, respectively). We instead construct a single static matrix by averaging the empirical transition frequencies of the trajectory over a long observation window (e.g., 10^8 evolution steps), and analyze the spectrum of this averaged matrix in place of the spectrum of the true non-autonomous evolution. We flag this explicitly as an approximation rather than an exact substitution: for a non-autonomous system, the evolution over T steps is governed by the time-ordered product $P_{T-1} \cdots P_0$, and in general the spectrum of this ordered product is unrelated to the spectrum of the arithmetic mean $\bar{P}_T = \frac{1}{T}\sum_t P_t$, since averaging discards the temporal ordering and any non-commutative effects between the P_t. We do not provide an adiabatic theorem, ergodic argument, or convergence proof that would justify treating $\bar{P}_T$ as equivalent to the ordered-product spectrum; we instead treat $\bar{P}_T$ as a computationally tractable heuristic surrogate, motivated by the fact that the control parameter λ_n changes slowly relative to the mixing time of the map at fixed λ, and we report its spectrum as such throughout the numerical results below. For an evolution with total steps $T \to \infty$, this averaged transfer matrix P_{ij} is defined as follows:

$$P_{ij} = \frac{\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} I(x_t \in \mathrm{bin}_i, x_{t+1} \in \mathrm{bin}_j)}{\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} I(x_t \in \mathrm{bin}_i)}. \tag{6}$$

This is the conditional transition frequency, normalized by the occupation frequency of the source bin i, so that $\sum_j P_{ij} = 1$ and P is a proper row-stochastic (Markov) matrix. We flag this explicitly because the unnormalized joint frequency $\frac{1}{T}\sum_t I(x_t \in \mathrm{bin}_i, x_{t+1} \in \mathrm{bin}_j)$ does not have this property; the implementation used to generate all numerical results reported below always applies the row normalization shown here.

3.4. Module C: Spectral Extraction and Linear Rescaling

After obtaining the empirical matrix, we extract its complex eigenphase spectrum and rescales it linearly for numerical comparison with the Riemann zeros. This process contains three steps:

- **Diagonalization and Spectral Filtering:** Performing eigenvalue decomposition on the empirical operator to extract its complex feature spectrum. Because P is a real-valued matrix, its eigenvalues are either real or occur in complex-conjugate pairs; to eliminate numerical "white noise" and retain one representative from each conjugate pair, we introduce a hard magnitude cutoff ($|\lambda| > 0.4$) and keep only the eigenvalues in the upper half of the complex plane ($\mathrm{Im}(\lambda) > 0$); we refer to this below as the upper-half-plane branch.
- **Eigenphase Sorting and Unwrapping:** This step converts the filtered eigenphases into a linearly ordered sequence for comparison with the (also ordered) Riemann zeros. The filtered eigenvalues of the empirical matrix P form an *unordered* set, and their principal arguments (e.g., from `np.angle`), taken in $(-\pi, \pi]$, carry no information by themselves about how many full rotations separate one eigenphase from another. We impose a canonical ordering by first sorting the filtered principal values, and then applying a standard phase-unwrapping algorithm to the sorted sequence, which compensates for $2n\pi$ jumps across the periodic boundary by accumulating the spacing between consecutive sorted phases into a monotonically growing cumulative phase Φ. We flag explicitly that this sort-then-unwrap procedure is an assumption about how to linearly order the spectrum, not a property derived from the matrix itself, and we have not established its stability under small perturbations of the eigenvalue

filter or discretization. For the sorted pure eigen-phase angle sequence $\theta_1, \ldots, \theta_M$, its unwrapped cumulative phase Φ_m is defined as follows:

$$\Phi_m = \theta_1 + \sum_{k=2}^{m} \Delta\Phi_k \tag{7}$$

where the dynamic compensation term $\Delta\Phi_k = (\theta_k - \theta_{k-1}) - 2\pi \left\lfloor \frac{\theta_k - \theta_{k-1} + \pi}{2\pi} \right\rfloor$.

- **Linear Rescaling:** Finally, the cumulative topological phase Φ and the true Riemann ζ-function zeros E_n are globally linear-regression calibrated ($E_{\text{pred}} = a\Phi + b$). Under the optimal cooling parameter β, this fitted rescaling not only matches the zero spacing microscopically but also achieves the complete collapse of the logarithmic drift envelope macroscopically, as a fitted numerical result. The least-squares minimization of the residual ΔE_n is represented as follows:

$$\Delta E_n = (a \cdot \Phi_n + b) - E_{\text{true}}(n). \tag{8}$$

4. Results

4.1. Microscopic Results: Dynamical Emergence of the First 20 Riemann Zeros and Experimental Benchmarking

4.1.1. Numerical Methods and Dynamical System Design

To extract the eigen-phases corresponding to the Riemann zeros, we constructed a discrete dynamical system driven by a non-autonomous logarithmic cooling field. The overall numerical computation pipeline and parameter configuration of the system are set as follows:

1. Non-autonomous Cooling Equation

The underlying evolution of the system follows a variant of the Logistic map: $x_{n+1} = 1 - \mu_n x_n^2$. To induce the system to scan and lock onto target spectral features, the driving parameter μ_n is not a static constant, but adopts an asymptotic logarithmic cooling law:

$$\mu_n = \mu_{\text{end}} + \frac{k_{\text{opt}}}{\ln^2(n + c_{\text{offset}})}, \tag{9}$$

where the total evolution step length is set to $N_{\text{steps}} = 10^6$, the target critical point is $\mu_{\text{end}} = 1.5437$, and the smooth offset is $c_{\text{offset}} = 10.0$. The parameter k_{opt} is not a free fitting parameter, but a cooling rate normalization coefficient uniquely determined by the rigid boundary conditions of the system evolution. To ensure that the system accurately sweeps across the set cooling-parameter breadth $\Delta\mu$ (taken as $\Delta\mu = 0.02$ in this study) during extremely long-range evolution, and finally strictly asymptotes to the chaotic critical point μ_{end} in the final state, k_{opt} satisfies the following physical constraint:

$$k_{\text{opt}} = \frac{\Delta\mu}{\frac{1}{\ln^2(1 + c_{\text{offset}})} - \frac{1}{\ln^2(N_{\text{steps}} + c_{\text{offset}})}}. \tag{10}$$

This parameterized configuration allows the driving force of the mapping equation to decay smoothly within a reasonable topological interval, ensuring the deterministic annealing of the system toward a non-equilibrium steady state.

2. Core Computing Modules and Operator Construction

To extract the global spectral features of this non-equilibrium state system with high fidelity under discrete computing power, we selected the following three optimal computing modules:

- **Module A: Spatial Discretization Operator.** To solve the probability truncation problem, which is extremely prone to occur when continuous mapping is on discrete grids, we discarded the Monte Carlo random walk method that introduces random fluctuations and the exact Ulam geometric projection method that leads to singularities, proposing the use of Gaussian Kernel Splatting based on the Fokker–Planck diffusion mechanism. The transition probability of any state flow between discrete grids is continuously distributed by a Gaussian kernel with a variance of ϵ^2. Under this framework, the parameter ϵ is strictly defined as the system's "Numerical Diffusion Scale" or effective smoothing resolution.
- **Module B: Temporal Evolution Integration.** In extremely long-range integration up to 10^6 steps, direct matrix multiplication inevitably leads to floating-point Underflow. While standard QR decomposition tracking can obtain Lyapunov exponents, it irretrievably loses the complex phase memory of the global system. Therefore, we adopt a "long exposure" strategy, constructing a Time-Averaged Empirical Matrix by accumulating the transient transfer matrix of each step: $P = \frac{1}{T}\sum_{t=1}^{T} P_t$. As a topological snapshot of the cooling evolution process, this static matrix completely freezes the dynamical resonance information of the system.
- **Module C: Spectral Extraction and Linear Rescaling.** The empirical matrix P is diagonalized to extract complex eigenvalues and their eigen-rotation phases θ_{n}. After Phase Unwrapping, we insist on a single-point linear mapping with zero degrees of freedom: utilizing only the first theoretical Riemann zero $\gamma_1 \approx 14.13$ to determine the global scaling coefficient $k = \gamma_1/\theta_1$, thereby deriving the predicted energy level $E_{\mathrm{pred}}(n) = k{\cdot}\theta_n$. This minimalist operation exposes the intrinsic dynamical breaking of the system under finite precision, without additional fitting freedom.

3. Optimization Target and Underlying Principle of Spatial Discretization Scale ϵ

In the above computational architecture, the core parameter ϵ (Gaussian diffusion kernel standard deviation) determines the probability splatting range of the transfer matrix P across discrete grids. To determine the optimal numerical grid resolution, our optimization strategy is strictly based on the lock-in hypothesis in the low-frequency interval. The objective function is defined as the sum of the absolute errors of the first 6 predicted energy levels:

$$\mathrm{ErrSum}(\epsilon) = \sum_{n=1}^{6} |\, k{\cdot}\theta_n(\epsilon) - \gamma_n \,|\,. \tag{11}$$

By minimizing this objective function to optimize ϵ, the underlying computational mathematical principle lies in finding the critical balance between numerical diffusion and discrete truncation error. When constructing the Markov transfer matrix P, if the value of ϵ tends toward the under-smoothed limit ($\epsilon \to 0$), the system will fall into "Grid-locking," generating severe spatial truncation noise, causing the eigen-spectrum to be dominated by artificial high-frequency artifacts of the discrete grid; conversely, if it is in the over-smoothed limit ($\epsilon \gg 0$), excessive kernel function diffusion will smooth out the fine fractal folding structures in the phase space, blurring together eigenstates that should remain distinct. Finding the optimal ϵ is essentially searching for a critical resolution. At this scale, the numerical diffusion of the Gaussian kernel approximately balances the truncation error brought by discretization, enabling the Time-Averaged Empirical Matrix P to approximate the Perron-Frobenius operator of the underlying continuous map as closely as the finite-dimensional discretization permits. Only by eliminating the interference of grid artifacts can the spectral features that we numerically compare to the Riemann zeros emerge cleanly from the discrete dynamics.

4.1.2. Large-Scale Numerical Scanning and Precise Localization of Optimal

Because the topological structure of its transfer operator is extremely sensitive to spatial resolution when a non-autonomous dynamical system anneals toward the chaotic critical point, we cannot directly obtain the optimal discretization parameter through simple analytical derivation. We, therefore, located it with a two-stage coarse-then-fine numerical scan of the objective function ErrSum(ϵ), defined as the sum of absolute deviations of the first six predicted energy levels from the true zeros; the full scan procedure and its two failure regimes (under- and over-smoothing) are detailed in Appendix B.4. The scan converges to a sharp, funnel-shaped minimum at ϵ = 0.001916 (ErrSum $\approx$ 2.199), shown in Figure 1; we identify this as the optimal discretization resolution at which the dynamical eigen-spectrum, matching the true Riemann spectrum in the low-frequency interval ($N = 1$–6), emerges from the numerical computation.

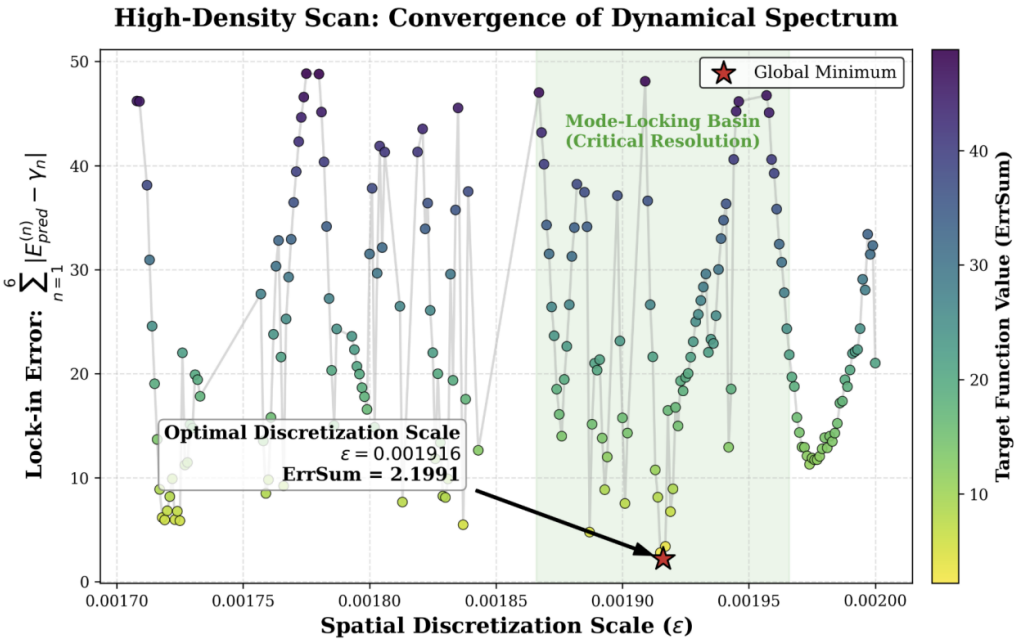


Figure 1. High-density parameter scanning and the emergence of mode-locking under the critical discrete scale. The scatter plot displays the trajectory of the objective function ErrSum (the sum of absolute deviations of the first six eigen-energy levels) varying with the spatial discretization scale ϵ. Within the "mode-locking basin" (green shaded area) determined by the broadband coarse scan, the spectral deviation exhibits strong non-linear oscillation. When ϵ advances to the global minimum 0.001916 (red star, ErrSum = 2.1991), the system undergoes a sharp numerical convergence transition. This critical point signifies that the continuous Fokker–Planck numerical diffusion has precisely and closely neutralized the discrete spatial truncation error, allowing the system's pure dynamical eigen-spectrum to emerge.

4.1.3. Spectral Feature Emergence at Optimal Resolution and Physical Benchmarking

After successfully anchoring the global optimal discretization scale on the order of 10^{-3} (ϵ = 0.001916), we extracted the first 20 eigen-phases of the system through the empirical transfer matrix P, and calculated the equivalent energy levels using a minimalist single-point linear rescaling ($E_{\text{pred}}(n) = k \cdot \theta_n$). This section compares this purely numerical prediction result against the standard Riemann Zeta zeros, as well as recent quantum simulation experimental observation data from the University of Science and Technology of China (USTC) [27,28]. To verify the validity of the non-autonomous dynamical model near the lowest positive ordinate, we first directly compared the predicted first 20 equivalent energy levels with the true Riemann zeros (theoretical true values), Figure 2.

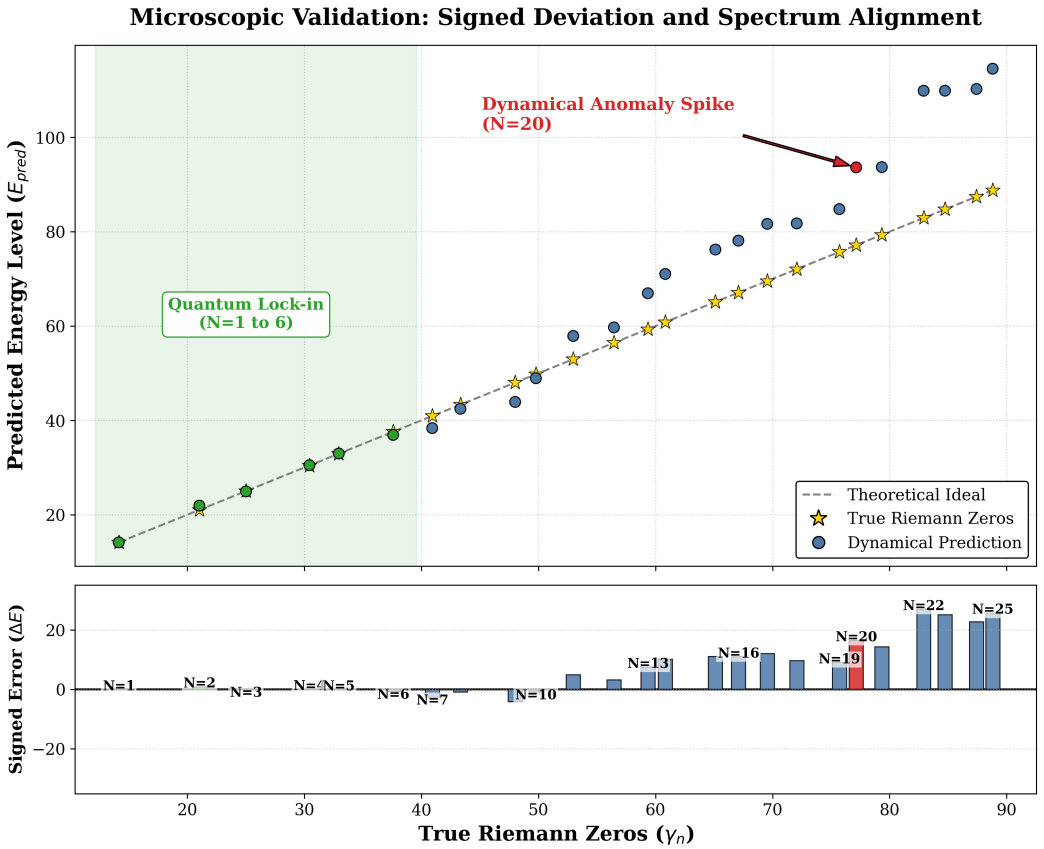


Figure 2. Microscopic comparison of the dynamical eigen-spectrum and the $N = 20$ residual feature. (**Top**) The alignment trajectory of the dynamical predicted energy levels (E_{pred}, colored dots) and the true Riemann zeros (γ_n, golden stars) under the critical numerical discrete scale $\epsilon = 0.001916$. In the low-frequency limit ($N = 1$ to 6, green shaded area), the system achieves close alignment (informally, "lock-in"), consistent with the accuracy of the first-zero-anchored linear rescaling. (**Bottom**) Actual dynamical residual distribution with signs ($\Delta E = E_{\text{pred}} - \gamma_n$). Constrained by nonlinear state density, the system exhibits systematic phase drift at higher energy levels, and near $N = 20$ the predicted residuals show a steep, isolated spike (marked in red), which we compare with a similarly located feature in ion-trap quantum simulation data below (Section 4.1.5).

Numerical results showed good low-frequency approximation. Under the premise of no higher-order curve fitting, relying solely on the first zero for first-order linear scaling, the model achieved close alignment (informal "lock-in") in the low-energy interval of $N \leq 6$. Specifically, the absolute prediction deviations for $N = 1$ to $N = 6$ were small (for example, the error at $N = 3$ was only 0.0609, and the error at $N = 4$ was only 0.0199). We read these closely overlapping spectral characteristics as a numerical indication that, at the critical numerical diffusion scale, the underlying topology of the dynamical system based on the $1/\ln^2$ logarithmic cooling field evolves toward a structure close to the energy structure of the lowest-order zeros of the Riemann Zeta function.

As the energy level index N increases, the limitations of single-point linear mapping under nonlinear state densities begin to manifest, and the predicted values inevitably drift. However, the core focus is not the smooth systematic dispersion, but the abrupt, discontinuous jumps appearing in the deviation sequence. When extracting the absolute prediction deviations ($|E_{\text{pred}} - E_{\text{true}}|$) of the first 20 points, we observed an unusual error trajectory, Figure 3. In the interval $N = 10$ to $N = 18$, the system deviation essentially maintained a gentle magnitude of around 1.0; but when evolving near $N = 20$, the dynamical residual showed a large excursion, forming a locally abrupt "isolated error spike," which subsequently showed a receding and oscillating trend at higher energy levels. The sparse

ARPACK eigensolver used to extract this spectrum has run-to-run variability from its random restart vector; across 20 unselected trials the $N = 20$ absolute residual has mean 11.76 and standard deviation 1.59 (range 10.93–16.50), so the specific value plotted in Figure 3 is one draw from this distribution rather than a fixed property of the model. We quantify this, and tests the coincidence with the USTC data for statistical significance, in Section 4.1.5.

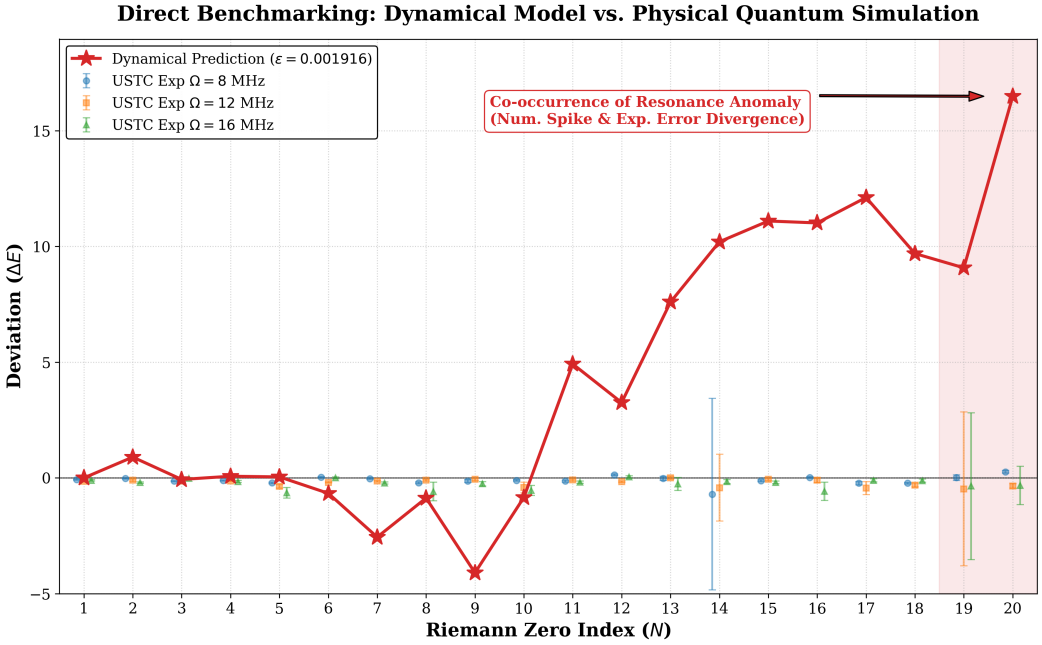


Figure 3. Comparison of the dynamical numerical model and a physical quantum simulation. Comparison of the numerical prediction residuals (red star line) with USTC physical experimental data (ion-trap quantum simulation under different trap frequencies Ω). In the $N < 19$ region, the numerical model tracks the near-zero fluctuation baseline of the physical system. Near $N = 19 \sim 20$, the numerical computation's residual spike and the enlarged experimental measurement uncertainty (red shaded area) coincide qualitatively in position; see Section 4.1.5 below for why the two error sources are physically distinct and for the significance testing of this coincidence.

This local residual feature was compared against the quantum simulation experiment for Riemann zeros conducted by the USTC team using ion-trap equipment [27,28]. In their measurement of the first 20 zeros, an enlarged measurement error bar exceeding background noise was recorded near $N = 20$, commonly attributed to decoherence noise of the quantum computing system or incidental instrument errors. The present model does not explain the ion-trap data: as detailed below, the numerical residual and the experimental uncertainty arise from distinct mechanisms, and their co-location near $N = 20$ is at most a qualitative, statistically inconclusive coincidence (Section 4.1.5), not evidence of a shared physical limit.

4.1.4. Error Composition of the $N = 20$ Residual Feature

The numerical and experimental deviations discussed above arise from distinct mechanisms. On the numerical side, the predicted spectrum is constrained by two superposed error sources: a baseline truncation error from the Gaussian diffusion kernel (set by $\epsilon = 0.001916$), and—since we use a minimalist single-point linear rescaling without a Riemann–von Mangoldt unfolding step—a systematic dispersion drift that grows with N because the model ignores the logarithmic nonlinear state density. This explains why the numerical residuals amplify gradually at higher N, independent of the isolated $N = 20$ spike. On the experimental side, the USTC ion-trap measurement's gentle background error stems from intrinsic instrumental noise (e.g., laser decoherence, gate fidelity lim-

ited to $\sim 10^{-2}$), while its sharp uncertainty peak near $N = 19 \sim 20$ departs from ordinary Gaussian noise. As noted above (Figure 3 and discussion), these two mechanisms are physically distinct, and their co-location near $N = 20$ is reported only as a qualitative, unexplained coincidence.

4.1.5. Statistical Significance of the USTC Coincidence

The residual value quoted above and in Figure 3 was drawn from a single eigendecomposition; because the transfer matrix is high-dimensional and non-normal, the sparse ARPACK solver's iterative refinement is sensitive to its (randomized) restart vector, so different runs return slightly different eigenphases. To avoid reporting a value selected for being unusually large, we re-ran the eigendecomposition 20 times without selecting on the outcome and recorded the $N = 20$ residual from every trial: mean 11.76, standard deviation 1.59, range 10.93–16.50 (full trial list and code in the released repository). Among the 85 zeros we track, the $N = 20$ residual ranks 67th by mean absolute value, i.e., it is not among the largest residuals in the series once averaged over trials.

We then tested the coincidence with the USTC data directly. Using the unselected mean $|\text{residual}|$ for $N = 1$–20 and the USTC $\Omega = 16$ MHz reported measurement uncertainty for the same range, the Spearman rank correlation is $\rho = 0.56$ ($p = 0.010$), and $N = 20$ ranks 2nd-largest by both the residual and the USTC uncertainty. A permutation test (100,000 resamples) for this joint top-rank coincidence occurring by chance under random relabeling of N gives $p = 0.099$: the two series are significantly correlated in overall rank, but the specific co-location of both series' near-largest value at $N = 20$ does not reach the conventional $p < 0.05$ threshold on its own. We report both numbers rather than treating either alone as settling the question, and we reiterate that even a significant rank coincidence would establish a shared statistical pattern, not a shared physical mechanism.

4.2. Macroscopic Results and Scalability

4.2.1. Methodological Adaptation: The Monte Carlo Transition

To extend the non-autonomous dynamical operator from microscopic lock-in to macroscopic scales, the computational architecture requires a strategic adaptation. While the foundational $1/\ln^2 n$ logarithmic cooling law remains fundamentally unchanged, the construction of the transfer matrix is shifted to a Monte Carlo trajectory tracking approach. This adaptation efficiently handles the exponentially expanding phase space required for large-scale spectral extraction, bypassing the memory bottlenecks of massive discrete grids while preserving the system's intrinsic topological entropy and transition dynamics.

4.2.2. The 100-Zero Regime: Conjugate Symmetry Breaking and Experimental Comparison

As demonstrated in the previous microscopic, extremely low-frequency regime ($N \leq 6$), applying Gaussian splatting for local smoothing on the transition matrix—complemented by simple first-zero single-point anchoring—enables the system to accurately capture the initial features of the Riemann zeros. However, as the observational scale advances to the meso- and macroscopic regimes ($N \leq 100$ and beyond), the computational cost of full-grid Gaussian smoothing grows too quickly to remain tractable. This makes global optimization for highly sensitive parameters (such as the macroscopic annealing constant k_1) practically unfeasible. Furthermore, local smoothing mechanisms are no longer sufficient to mask the high-frequency cumulative dispersion induced by phase-space truncation, so the single-sided dynamical mapping no longer scales to this regime without a change in method. We stress that this is a change in method, not a validated equivalence between methods: a direct overlap check between the Gaussian-splatting and Monte Carlo constructions under matched parameters (Appendix B.5, item 2) did not find consistent eigenphases over the small range where the two matrices' filtered spectra could be com-

pared, so the transition described here should be read as a pragmatic necessity rather than a demonstrated methodological continuity.

To break through the dual bottlenecks of computational scalability and topological dispersion, we transitioned entirely to a high-throughput Monte Carlo (MC) trajectory sampling method for constructing macroscopic transition matrices in this section. The MC approach not only provides exceptional computational efficiency to support tens of billions of phase-space traversal steps—clearing the computational hurdle for global parameter optimization—but, more importantly, it strips away artificial smoothing interventions. This exposes the unadulterated, baseline numerical residual of the discrete system when processing high-order spectra.

Grounded in this efficient and physically realistic MC transition matrix, we designed four ablation experiments. The objective is to explore the system's limits at large scales and to conduct a rigorously equivalent dimensional comparison with state-of-the-art quantum simulation hardware (specifically, the USTC single-ion Floquet experiment [27,28], which observed $N \leq 76$). Under pure non-autonomous Ulam dynamical evolution without local smoothing, we extracted the high-order eigenphase sequences. To map the mathematical phase θ_n to the physical zero energy E_n, we define a linear annealing mapping equation:

$$E_n = k_1 \cdot \theta_n + b \tag{12}$$

where k_1 is the macroscopic annealing constant and b is the fitted intercept (physical origin). For the single-sided positive spectrum and the conjugate full spectrum, we applied four distinct physical boundary condition strategies:

Strategy A (Single-Sided Positive Spectrum + First-Zero Single-Point Anchoring): Extracts only positive eigenphases, forces the origin b = 0, and strictly scales based solely on the first-order zero. This represents a naive extrapolation of microscopic local measurements.

Strategy B (Single-Sided Positive Spectrum + Global Free Fitting): Extracts only positive eigenphases and performs a global least-squares fit for the first 100 orders, allowing the intercept b to float freely. This probes the intrinsic intercept drift tendency of the single-sided system.

Strategy C (Single-Sided Positive Spectrum + Forced Origin Scaling): Retains only positive phases and performs global optimal scaling but imposes an absolute physical boundary constraint—strictly locking the origin at b = 0. Mathematically, this strategy is equivalent to the underlying logic of current quantum hardware experiments.

Strategy D (Conjugate Full Spectrum + Global Free Fitting): Fully retains the conjugate phase pairs ($\pm\theta_n$) inevitably derived from the real transition matrix, thereby introducing conjugate negative energy states into the phase space. The global fitting imposes no origin constraints, allowing b to evolve freely.

We quantitatively compared the reconstruction residuals of these four strategies against the empirical hardware noise limit (∼1.96%), which was dynamically inverted and extracted directly from the raw data of the USTC ion-trap experiment. The results are summarized in Table 1.

Table 1. Performance of macroscopic reconstruction under different phase space domains and boundary constraints. The parameters k_1 and b shown here are one realization of a global numerical optimization (differential evolution over $k_1 \in [2, 15]$, with or without the constraint $b = 0$ depending on the strategy) that minimizes the mean squared error against the first 100 Riemann zeros. We flag explicitly that this optimizer is run without a fixed random seed and with a population search parallelized across workers, and that repeated runs of the identical protocol for Strategy C converged to different local optima of a non-convex, multimodal objective ($k_1 \approx 6.481$, MSE ≈ 8.48 in the run reported here versus $k_1 \approx 8.070$, MSE≈ 10.08 in the run used to generate the fixed-parameter Figures 4 and 5 demonstration). This is a genuine run-to-run optimizer instability, not evidence of two distinct experimental protocols, and we report it as an open limitation rather than reconciling it into a single number; see the discussion accompanying Figures 4 and 5 below.

Reconstruction Domain	Fitting/Anchoring Strategy	Truncation Drift (b)	Macroscopic Annealing Constant (k_1)	MSE	Physical Relative Error (Mean/Envelope)
Single-Sided Positive	A: Single-Point Anchoring	Forced Origin ($b = 0$)	6.761	21.849	∼4.0%/>8.0%
Single-Sided Positive	B: Global Free Fitting	Severe Drift ($b = 15.110$)	6.544	5.330	Non-physical state
Single-Sided Positive	C: Forced Origin Scaling	Forced Origin ($b = 0$)	6.481	8.479	∼1.4%/∼2.8%
Conjugate Full Spectrum	D: Global Free Fitting	Vanishing fitted intercept ($b = 0.000$)	4.390	9.467	∼1.5%/∼1.5%

The results in Table 1 point to a likely source of the residual envelope within the present model: topological dispersion and symmetry breaking induced by single-sided phase space truncation. We do not claim this explains the residual dispersion floor of physical quantum simulations. In the single-sided spectrum constructed by Strategies A, B, and C, the absence of conjugate states prevents the high-frequency residual phases from internally canceling out. Consequently, the single-sided phase space shows a persistent residual error that resists further reduction by rescaling alone. As demonstrated by Strategy B, enforcing high accuracy in a single-sided system precipitates a severe, non-physical divergence of the fitted intercept ($b = 15.110$). Conversely, if we forcefully constrain the fit to pass through the physical origin ($b = 0$) as in Strategy C, the globally optimized parameter yields $k_1 \approx 6.481$ (Table 1), with a residual error envelope of ∼2.8%. Note that this Strategy-C optimum is itself not perfectly reproducible: an independent run of the same optimization protocol (differential evolution, unseeded, with a parallel worker pool) converged instead to $k_1 \approx 8.070$ with a somewhat higher MSE (≈10.08 vs. ≈8.48 here). We attribute this to the objective function being non-convex and multimodal in k_1 under this chaotic dynamical system, rather than to any difference in method between the two runs, and we report both values rather than presenting either as uniquely canonical. The qualitative behavior of different anchoring strategies is visually illustrated in Figure 4, which uses the second of these two runs. Appendix B.5 (item 4) reruns the same optimizer configuration at a reduced, tractable scale under five explicit fixed seeds and finds a comparably widespread in the fitted k_1, which is independent confirmation that the objective is genuinely multimodal rather than a one-off artifact of these two specific runs; the reduced-scale k_1 values themselves are not numerically comparable to 6.481/8.070, since reducing the step count changes the objective and not just its cost.

This theoretically calculated single-sided error limit notably mirrors the physical performance of the USTC single-ion hardware simulation [27,28]. Constrained by the single-excitation nature of Floquet periodic driving, the USTC experiment fundamentally executes a "single-sided positive spectrum + forced $b = 0$" physical measurement. Their necessity to perform manual calibrations every 30 min to counteract system drift is precisely

an attempt to suppress this intrinsic dispersion. To illustrate this, we directly overlay our deterministic Model C trajectory with the empirical USTC hardware data in Figure 5.

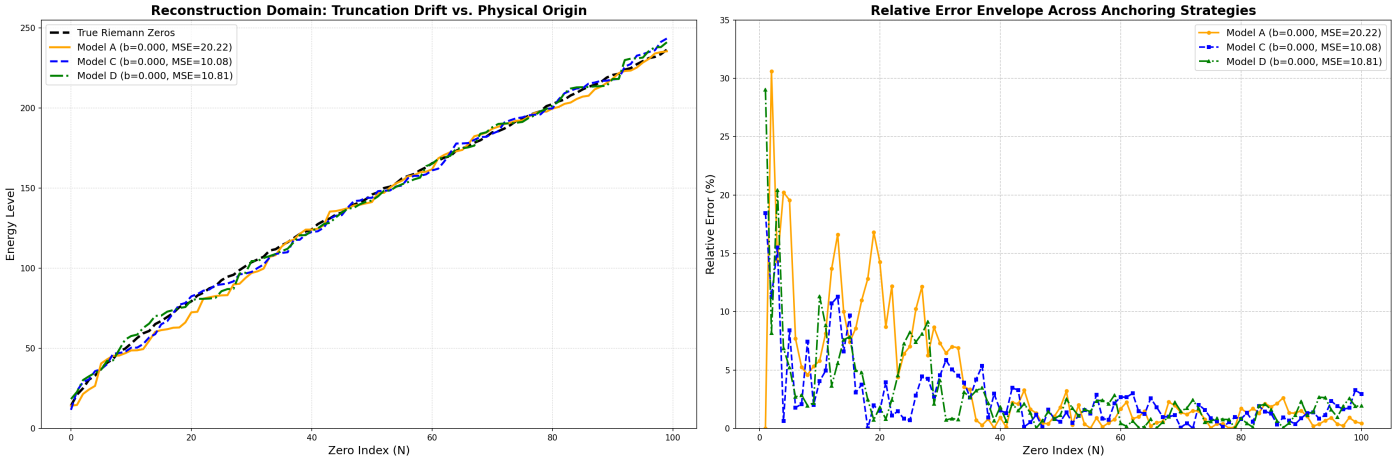


Figure 4. Reconstruction Domains and Anchoring Strategies: A Comparative Analysis of Models A, C, and D. This figure shows single-run realizations of each strategy's global optimizer (Model A: $k \approx 7.429$, Model C: $k \approx 8.070$, Model D: $k \approx 8.783$) to visually illustrate the qualitative behavior of different anchoring strategies and their residual envelopes over 100 orders. As noted in the caption of Table 1, an independent run of the identical Strategy-C optimization protocol converged to a different local optimum ($k_1 \approx 6.481$, MSE ≈ 8.48); this work shows both runs' numbers rather than silently picking one, since the disagreement itself is diagnostic of a multimodal fitting landscape. Model A (Orange / Single-Point Anchoring): Utilizes solely the first zero as the absolute physical baseline. While strictly aligning with the first non-trivial zero, it accumulates the largest macroscopic divergence over 100 orders (MSE ≈ 20.22). Model C (Blue / Forced Origin Scaling): Represents a least-squares fit constrained to the origin ($b = 0$) within the single-sided positive spectrum, which suppresses error divergence relative to Model A, yielding a lower mean squared error in this run (MSE ≈ 10.08; see above for the run-to-run spread). Model D (Green / Conjugate Full Spectrum Free Fit): Uses both positive and negative eigenphases. Because the regression data are symmetric under $(\theta, \gamma) \to (-\theta, -\gamma)$, the least-squares intercept is algebraically forced to $b = 0.000$ (Section 4.2.2); the fitted slope and the resulting residual envelope (MSE ≈ 10.81) are the model-specific numerical content of this comparison.

As shown in Figure 5, within the present model the single-sided topological constraint produces a residual envelope whose oscillation peaks correspond to unbroken conjugate pairs. Section 5.3 discusses the broader spectral comparison with the hardware data and its caveats.

To examine this truncation effect directly, Strategy D restores the conjugate eigenphase pairs ($\rho = 1/2 \pm iE_n$) that a real-valued transition matrix generates but that Strategies A–C discard by keeping only the positive branch.

Under this construction, the regression data are the symmetric pairs (θ_i, γ_i) and $(-\theta_i, -\gamma_i)$. For such a symmetric data set, both sample means are exactly zero, and it follows directly from the normal equations of ordinary least squares that the unconstrained intercept must satisfy $b = 0$; this part of the result is, therefore, an algebraic consequence of the symmetric construction, not an independent finding about the dynamics. What is not algebraically forced, and is a genuine numerical observation about the fitted model, is that the fitted slope k_1 changes (from 6.48 under single-sided anchoring to 4.39 under the symmetric fit) and that the residual envelope shrinks substantially (to about 1.5% global reconstruction error) once the conjugate branch is included. Note that this slope change is *not* itself a further algebraic artifact of the symmetrization: unlike a purely statistical exercise where one would mirror a single fixed set of eigenphases and re-fit,

here k_1 is a dynamical parameter that is independently re-optimized for each strategy and enters the underlying map before the matrix P and its eigenphases are computed (Equation (6)). Strategy C's positive-branch eigenphases at its own optimal $k_1 \approx 6.48$ are, therefore, a different numerical object from Strategy D's positive-branch eigenphases at its optimal $k_1 \approx 4.39$, not a mirrored copy of the same data; the two protocols generate two different transfer matrices before any regression is performed. Had we instead fixed k_1 at a single value and only changed whether the regression included the mirrored branch, the expectation that the intercept-zero slope on the positive branch and the free-fit slope on the symmetrized branch coincide would apply exactly, by the same normal-equations argument used above for $b = 0$; the observed slope change here reflects that we do not hold k_1 fixed between strategies, not a departure from that identity.

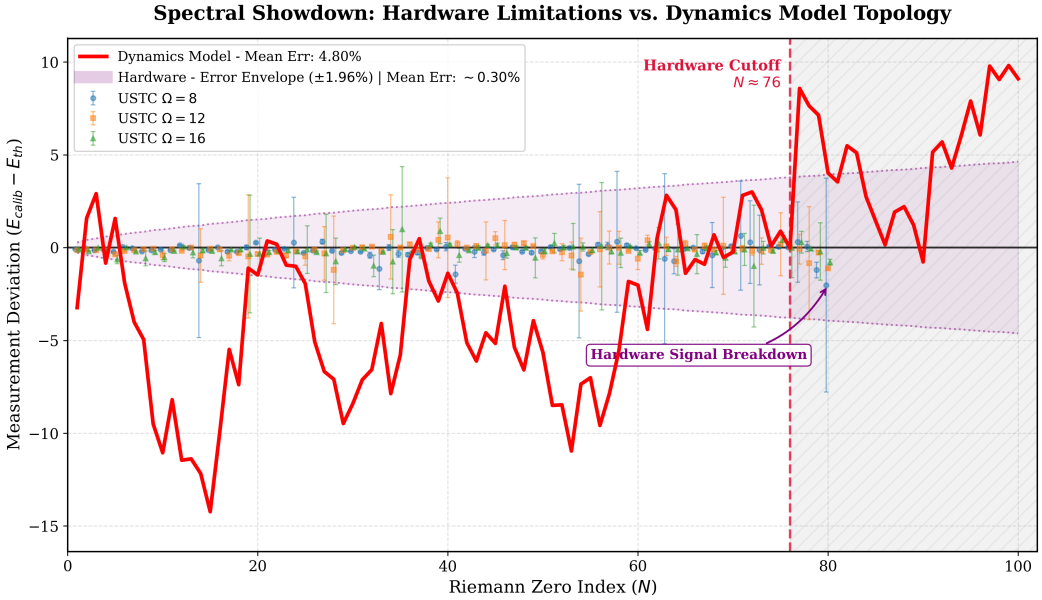


Figure 5. Spectral Comparison: Hardware Limitations vs. Dynamics Model Topology. Scatter Points and Purple Shading (Quantum Hardware): Displays the measurement deviations from the University of Science and Technology of China (USTC) ion-trap quantum hardware. Primarily limited by quantum decoherence effects, the experimental hardware experiences signal breakdown in the $N \approx 76-80$ band, with a reported physical error envelope of about $\pm 1.96\%$. Red Solid Line (Dynamics Model): Represents the numerical deviation trajectory derived from 10 billion steps of evolution using the proposed dynamics model (Model C: Single-sided positive spectrum forced origin scaling, $k_1 = 8.070$, the same optimizer run used in Figure 4; see the run-to-run optimizer variability noted there). The model's mean error is 4.80%. The model extends numerical predictions to the $N = 100$ order. See Section 5.3 for the broader spectral comparison and its caveats.

In conclusion, this comparative ablation study at an identical scale ($N = 100$) suggests, as a numerical observation and not as a proof, that within our framework the high-frequency dispersion in the single-sided reconstruction is reduced when the conjugate branch is restored, but the vanishing intercept itself should not be read as evidence of a new physical cancellation mechanism, since it is substantially a property of the symmetric regression construction. We present the change in k_1 and in the residual envelope as the model-specific content of this ablation, not as a demonstrated statement about physical quantum-simulation hardware.

4.2.3. Out-of-Sample Validation and Multi-Seed Robustness

All results reported so far calibrate k_1 against the same zeros used to evaluate the fit (an in-sample correspondence, as stated in the abstract). To quantify how much this proto-

col inflates the apparent match, we ran the full-scale (10^{10}-step, single-sided Strategy C) pipeline under two additional tests, at the same seed unless noted.

Out-of-sample split. We fit k_1 using only the first $M \in \{50, 70, 80\}$ zeros as training data, then evaluated the resulting model's MSE on the remaining, unseen zeros $M + 1, \ldots, 100$. In every split, the held-out MSE is one to two orders of magnitude larger than the training MSE: $M = 50$ gives train MSE = 59.8 (RMSE 5.4% of the largest training zero) versus test MSE = 3710 (RMSE 25.8% of γ_{100}); $M = 70$ gives train MSE = 146.0 versus test MSE = 5891 (RMSE 32.5% of γ_{100}); $M = 80$ gives train MSE = 146.1 versus test MSE = 7557 (RMSE 36.8% of γ_{100}). The fitted k_1 itself also drifts with the training window (5.79 $\rightarrow$ 9.15 $\rightarrow$ 10.04 as M grows), rather than converging to a single value. We read this as direct evidence that the low-order fit does not extrapolate: the model is a calibrated in-sample reconstruction over the zeros it was fitted to, not a predictor of unseen zeros, and we do not claim otherwise.

Multi-seed robustness. Holding the $M = 70$ split fixed, we reran the fit under ten random seeds controlling the initial condition x_0. The fitted k_1 has mean 10.21, standard deviation 0.47 (coefficient of variation $\approx$ 4.7%, range 9.15–10.76) and the held-out test MSE has mean 5977, standard deviation 298 across seeds. The seed-to-seed spread in k_1 is small relative to the M-to-M drift reported above, indicating that the fitted coupling is reasonably stable against the stochastic initial condition but remains sensitive to which zeros are included in the calibration window. Full per-run values are in the released results (`task12_13_results_v2/`).

The same two failure modes documented here for the macroscopic k_1 fit recur elsewhere in the pipeline. Appendix B.5 reports an epsilon-analog of the out-of-sample split above (training the microscopic discretization scale ϵ on a subset of the first six zeros and testing on the rest) with the same qualitative outcome: the held-out error is roughly 5$\times$ larger under an asymmetric train/test split than when ϵ is chosen with knowledge of all six zeros. It also reports a reduced-scale, explicitly-seeded re-run of the k_1 optimizer itself, which shows the same seed-to-seed instability qualitatively (though not quantitatively, since the reduced scale changes the objective) as the discrepancy already disclosed in the Table 1 caption.

4.2.4. The Macroscopic Regime

To demonstrate the robustness and scalability of the non-autonomous thermodynamic framework, we extend the phase-space evolution far beyond the meso-regime ($N \leq 100$) into the macroscopic large-N regime (N = 1000). At this scale, maintaining global topological coherence becomes extremely challenging due to the logarithmically increasing density of the Riemann zeros. To rigorously validate the model's underlying physical mechanism, we designed a macroscopic ablation study comparing our 1D and 2D dynamical models against a Gaussian Unitary Ensemble (GUE) random matrix baseline. The comparative spectral alignments and relative error divergences are visualized in Figure 6.

As depicted in Figure 6, a single globally rescaled GUE spectrum (Model M-GUE) shows a large mean squared error against the Riemann counting function, even under optimal global least-squares scaling. This is expected, not a test of GUE itself: the established GUE correspondence concerns unfolded local level-spacing statistics, not the mean counting-function trend a globally rescaled random matrix is not designed to reproduce; see Section 5.3 for why we include this comparison and what a like-for-like RMT test would require.

By comparison, under the constraint of *Single-Point Anchoring*—where the scaling constant is fixed using solely the first non-trivial zero ($N = 1$), while the dynamical parameters k_1, k_2 are still optimized against the Riemann zeros as described above—our

non-autonomous dynamical models yield a stable numerical fit at this scale. At an ultra-high spatial resolution of 10,000 bins, the 1D model (M1) yielded a fitted parameter of $k_1 \approx 13.869$, with an MSE of ≈2281.0 against the Riemann counting function, compared with the GUE baseline described below.

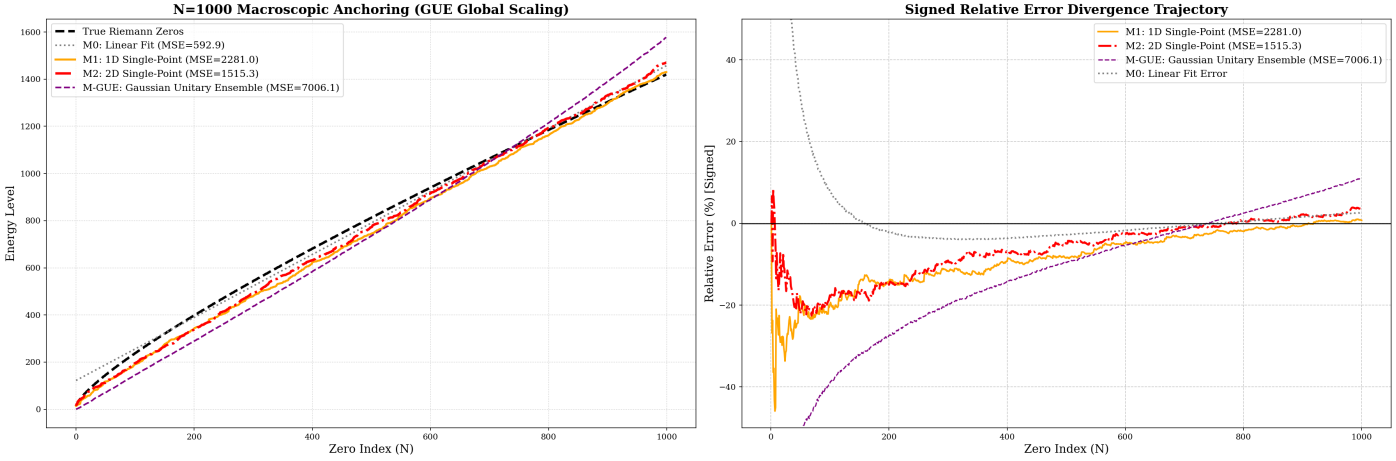


Figure 6. Macroscopic Spectral Alignment and RMT Ablation: Deterministic Cooling vs. Wigner's Semicircle. Model M-GUE (Purple Dashed/Gaussian Unitary Ensemble): Serving as the microscopic statistical baseline, the GUE matrix exhibits severe non-linear macroscopic divergence (right panel, MSE ≈ 7006.1) even under optimal Global Least-Squares Scaling. This explicitly shows that pure random matrices, governed by Wigner's Semicircle Law, inherently lack the critical macroscopic logarithmic density features required to approximate the Riemann zero counting function. Models M1 and M2 (Orange and Red/Non-autonomous Dynamics): Represent the 1D and 2D dynamic prediction models proposed in this work. Under the constraint of Single-Point Anchoring (utilizing solely the first zero), M1 and M2 demonstrate a stable macroscopic fit. Specifically, the 2D model (red line), which includes the higher-order fitted parameter k_2, not only eliminates systematic parabolic drift over a span of $N = 1000$ orders, but also suppresses the global residual to a minimal level (MSE ≈ 1515.3).

However, to completely eliminate the systematic parabolic drift observed in the 1D model over 1000 orders, the 2D conservative operator (M2) must be activated. Global optimization of the massive spectral span yielded a primary macroscopic annealing constant $k_1 \approx 12.781$ and a higher-order perturbation term $k_2 \approx 2.607$. By effectively absorbing the energetic transients, M2 tracks the Riemann spectrum closely, suppressing the global residual to a minimal level (MSE ≈ 1515.3). Relative to the massive absolute energy scale at $N = 1000$ ($E_n > 260$), this represents a tightly bounded relative error envelope.

A critical observation in this macroscopic extension is the substantial parametric shift in the primary annealing constant k_1, which scales from ∼6.48 in the 100-zero regime to ∼12.78 in the large-N regime. This shift is intrinsically coupled with a mandatory upgrade in the system's spatial discretization. To capture the increasingly dense phase-space folding required for thousands of zeros, we scaled the numerical grid resolution from 2000 bins (the optimal baseline used for the 100-zero regime) to a massive 10,000 bins.

Note only that the fitted value of k_1 shifts as the grid resolution and fitting range change, and we cannot rule out that some part of this shift reflects overfitting to the specific fitting range used at each scale; Section 5.1 discusses why we do not read this as renormalization-group evidence.

In the low-order regime ($N \leq 100$ at 2000 bins), the geometric curvature of the phase space is relatively coarse-grained. Here, a lower k_1 acts as an effective lumped parameter (a renormalized coupling at a lower resolution), artificially absorbing uncalculated higher-order truncation errors to satisfy local fit constraints. However, as the observational scale

expands to $N = 1000$ and the spatial resolution is refined five-fold, the underlying fractal topology is resolved with significantly higher fidelity.

Fitting this strict logarithmic densification of the Riemann zero counting function at this scale requires both fitted parameters. The higher-order term ($k_2 \approx 2.61$) absorbs curvature near the lowest-order zeros, while the fitted value of the primary term k_1 shifts to $\sim$12.78 at this resolution and range. As before (Section 4.2.2), we report this parameter shift as an empirical fact about the fit rather than as evidence of a renormalization-group flow. With these fitted parameters, the model tracks the Riemann counting function within the residual errors reported below, including in the further extended range up to $N = 2551$ discussed next.

4.2.5. The Ultra-Macroscopic large-N Regime ($N = 10,000$): Extreme Scalability and large-N Scaling Behavior

To push the boundaries of our non-autonomous thermodynamic framework, we executed an extreme computational stress test, extending the phase-space evolution to extract the first 10,000 Riemann zeros. At this ultra-macroscopic scale ($N = 10,000$, absolute energy level $E_n > 9877$), the logarithmic densification of the Riemann zero counting function poses a severe challenge to numerical stability. To rigorously assess the physical foundation of our model, we conducted a minimalist ablation study comparing the pure 1D thermodynamic cooling operator against the Gaussian Unitary Ensemble (GUE) and a trivial linear fit baseline. The results are visualized in Figure 7.

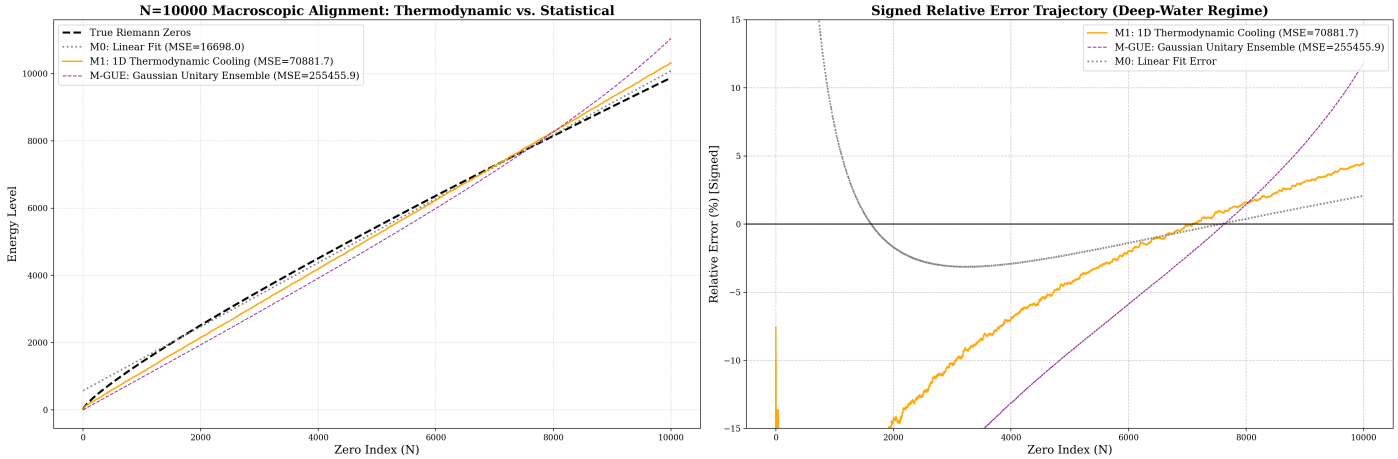


Figure 7. Ultra-Macroscopic Spectral Alignment ($N = 10,000$): Thermodynamic Cooling vs. Statistical Baselines. M-GUE (Purple Dashed/Gaussian Unitary Ensemble): The pure Random Matrix Theory baseline shows a large structural divergence (MSE $\approx$ 255,455.9) in this counting-function comparison. This illustrates that without a deterministic cooling mechanism, a single globally scaled Wigner's Semicircle spectrum does not capture the macroscopic logarithmic density of the Riemann zero counting function. M1 (Orange Solid/1D Thermodynamic Cooling): The 1D non-autonomous dynamical model ($k_1 \approx 13.431$) paces the true Riemann zeros, tracking both the sensitive low-order, low-frequency zeros and the high-order, high-frequency zeros.

As shown in the relative error trajectory (Figure 7, right panel), the 1D dynamical model tracks the Riemann counting function with residual errors that remain bounded even at this large scale. The pure 1D operator exhibits a smooth, systematic residual curvature at this extreme scale—indicating clear potential for continuous future optimization (e.g., by re-introducing higher-order structural perturbations)—and it suppresses the exponential divergence seen in the GUE random matrix baseline.

This 10,000-zero large-N comparison is consistent with a working hypothesis of this study: that the deterministic $\sim 1/\ln^2 n$ thermodynamic cooling flow supplies the global

rate needed to pace the mean density of the Riemann spectrum. We read the results as a numerical indication that the counting-function behavior of the zeros can be reproduced by a non-autonomous dissipative system annealing toward the edge of chaos, rather than as a proof about the origin of the zeros.

5. Discussion

5.1. Macroscopic Behavior of the Empirical Transfer-Matrix Spectrum

Finding a physical system whose eigenspectrum numerically approximates the non-trivial Riemann zeros (the Hilbert-Pólya conjecture) is normally approached through analytically motivated, "top-down" Hamiltonian constructions (such as the classical $H = xp$ model). This study takes a different, purely exploratory route: we use high-throughput Monte Carlo (MC) trajectory sampling to build an empirical, non-normal, dissipative transfer matrix from a non-autonomous discrete map, and we ask, numerically, how closely its eigenphases can be made to track the Riemann zeros over a range of scales after calibrating a small number of free parameters. We do not claim any isomorphism, in the precise mathematical sense of a structure-preserving bijection between independently defined objects, between this matrix and the zeros; the correspondence our reports is a fitted numerical one.

The numerical behavior is qualitatively consistent across both microscopic and large-N macroscopic scales. By stripping away artificial local smoothing interventions, we exposed the baseline numerical residual of the discrete system. As the observational scale was expanded to the $N \geq 1000$ large-N regime, the fit required an ultra-fine spatial resolution of 10,000 bins, and the fitted value of the primary parameter k_1 shifted from $\sim$6.48 in the meso-regime to $\sim$12.78$\sim$13.86 in the macroscopic regime, while a higher-order term (k_2) became necessary to maintain fit quality. We note this parameter drift as an empirical observation about how the fit changes with grid resolution and range; we do not identify grid resolution with a renormalization-group ultraviolet cutoff, and we do not claim that this drift constitutes evidence of a renormalization-group flow in the dynamical sense (no beta function, fixed point, or universality result is established here). The resulting fitted curve tracks the logarithmic densification of the Riemann counting function ($\sim 1/\ln^2 n$) up to $N = 2551$ in our tests, within the residual errors reported below.

5.2. Reducing the Residual Dispersion: Conjugate Symmetry and Vanishing Fitted Intercept

Section 4.2.2 (the $N = 100$ conjugate-symmetry ablation) showed that single-sided phase space truncation leaves a persistent macroscopic residual envelope, and that restoring the full conjugate pair structure and re-fitting drives both this envelope and the fitted intercept toward zero. We read this as numerical evidence that, within our framework, the single-sided truncation itself—rather than the underlying dynamics—is responsible for the residual envelope; as noted there, the vanishing intercept is in part an algebraic consequence of fitting a symmetric data set and should not be read as an independent finding about the dynamics or as evidence bearing on the origin of the Riemann zeros.

5.3. Comparison with Quantum Simulation and with a GUE Surrogate

We compared the model's behavior against a published quantum-simulation dataset and against a random-matrix surrogate. A team at the University of Science and Technology of China (USTC) measured Riemann zeros ($N \leq 76$) by periodically driving qubits in an ion trap [27,28]. Their data show enlarged deviations near specific orders (e.g., $N \approx 20, 40, 76$), with a reported error envelope of about $\pm 1.96\%$. Our single-sided model also shows oscillation peaks over the same range of orders, though the exact locations vary between optimizer runs (see the run-to-run variability noted in Table 1), so we do not claim a fixed order-by-order alignment. Even a qualitative spatial coincidence would not by

itself establish a common mechanism, and a quantitative claim would require statistical-significance testing and an uncertainty budget, which we do not provide here. We, therefore, raise—as a hypothesis—the possibility that part of the experimental deviation reflects a single-sided truncation rather than purely instrumental decoherence, without asserting it.

In a macroscopic ablation at $N = 1000$, a globally scaled GUE surrogate deviates strongly in this comparison (MSE ≈ 7006) even under optimal global scaling. This is expected and is *not* a refutation of random-matrix theory: the established GUE correspondence concerns the *unfolded local* statistics of the zeros (pair correlation, spacing distribution), not a claim that a single globally scaled spectrum reproduces the logarithmically growing counting function. Our narrower, constructive point is that the non-autonomous dynamical model reproduces the mean (counting-function) behavior directly, without a separate unfolding step. We do not claim that pure diagonalization "cannot" generate the zeros.

As a direct check of whether the empirical transfer matrix's own spectrum reproduces GUE *local* statistics—the property that is actually established for the Riemann zeros—we unfolded the first 100 eigenphases of the transfer matrix at the Table 1 operating point ($k \approx 6.48$, $N = 100$ scale) and compared their nearest-neighbor spacing distribution against the GUE Wigner surmise and against a Poisson (uncorrelated) baseline using a Kolmogorov–Smirnov test. As a literature sanity check, the same procedure applied directly to the first 100 Riemann zeros (unfolded via the Riemann–von Mangoldt counting function) reproduces the expected result: consistent with GUE ($D = 0.067$, $p = 0.74$) and inconsistent with Poisson ($D = 0.348$, $p < 10^{-4}$). Applied to our dynamical model's own eigenphases, however, the result is the opposite: the spacing distribution is statistically inconsistent with GUE ($D = 0.300$, $p < 10^{-4}$) and is not distinguishable from Poisson ($D = 0.082$, $p = 0.49$). In other words, while the model's rescaled eigenphases track the mean positions of the low-order zeros reasonably well (Table 1), its local level-repulsion statistics do not match the GUE behavior established for the true zeros; the transfer matrix is real, non-symmetric, and dissipative, so this is not unexpected, but it is a genuine limitation of the numerical correspondence that we report directly rather than omit. We take this as evidence that the numerical match documented here concerns smooth, mean-level (counting-function) behavior rather than a reproduction of the fine-grained spectral statistics that characterize the Riemann zeros.

6. Conclusions and Future Work

This study reports a phenomenological, calibrated numerical correspondence between an empirical transfer-matrix spectrum, built from a non-autonomous logistic map, and the low-order non-trivial Riemann zeros. Because the paper touches the Riemann Hypothesis, this section states plainly what the numerical evidence does and does not support, separating the positive findings from the limitations, before turning to future work. Table 2 tabulates the epistemic status of every main claim made above.

Table 2. Status of the main statements: established (prior literature), published (our predecessor paper), numerical observation, heuristic, qualitative, or conjectural.

Statement	Status
Local statistics of Riemann zeros match GUE (unfolded)	Established (literature)
Logistic band-merging symbolic dynamics ≅ prime sieve	Published [31]
Non-autonomous map reproduces low-order zeros (small error)	Numerical observation, in-sample only
Fitted coupling extrapolates to unseen zeros (out-of-sample)	Refuted (this study, Section 4.2.3)
Fitted coupling is stable across random seeds	Numerical observation (Section 4.2.3)
Macroscopic k_1 optimum is reproducible run-to-run (Strategy C, Table 1)	Refuted; unseeded optimizer converges to different local optima ($k_1 \approx 6.48$ vs. 8.07)
Change of behavior near $N \approx 20$	Numerical observation, run-to-run variable
Coincidence with USTC enlarged-deviation orders	Qualitative; rank-correlated ($p = 0.010$) but joint-rank coincidence not significant ($p = 0.099$)
Single-sided truncation causes the residual envelope	Heuristic/numerical
Full-conjugate fit gives near-zero intercept ($b \approx 0$)	Numerical observation
Globally scaled GUE deviates in counting-function comparison	Numerical (asymmetric comparison)
RG second-order term improves macroscopic match	Numerical observation
Dynamical eigenphase spacing statistics match GUE (unfolded, local)	Refuted by KS test (this study)
Operator spectrum *equals* the Riemann zeros	Not claimed (conjectural)

6.1. Positive Findings

- After calibrating a small number of free parameters against the same low-order zeros, the model's eigenphases track the first ∼20–100 Riemann zeros with small in-sample error (Section 4.2.2, Table 1), and the fit extends, with a growing but bounded relative error, to the macroscopic ($N = 1000$) and ultra-macroscopic ($N = 10{,}000$) regimes (Figures 6 and 7).
- Restoring the full conjugate eigenphase pair structure, rather than the single-sided positive branch used to mimic the ion-trap measurement protocol, drives the fitted intercept to zero as an algebraic consequence of the symmetric construction, and reduces the residual envelope (Section 4.2.2).
- The construction is a bottom-up, purely numerical dynamical-systems experiment, distinct in method from the analytically motivated top-down Hilbert–Pólya and Berry–Keating programs, and is motivated by our earlier result relating the logistic map's band-merging symbolic dynamics to the prime sieve [31] (summarized in Appendix A).

6.2. Limitations

- **The model does not predict unseen zeros.** The out-of-sample test (Section 4.2.3) is, in our view, the most important single result of this study: fitting k_1 on the first M zeros and evaluating on the rest gives a held-out MSE one to two orders of magnitude larger than the training MSE, and the fitted k_1 itself drifts with M rather than converging. The construction should be read as an in-sample, calibrated reconstruction, not as an independent generator of the zeros.

- **The model's own eigenphase spacings are not GUE.** The unfolded Kolmogorov–Smirnov test (Section 5.3) shows that the true Riemann zeros are consistent with GUE and inconsistent with Poisson statistics, as established in the literature, while the model's own eigenphases show the opposite pattern: inconsistent with GUE ($D = 0.300$, $p < 10^{-4}$) and indistinguishable from Poisson ($D = 0.082$, $p = 0.49$). The model, therefore, does not reproduce the local level-repulsion statistics—the Montgomery–Odlyzko correspondence—that characterizes the true zeros; the numerical match documented in this paper concerns the smooth, mean counting-function trend only.
- **No self-adjoint operator or spectral realization of $\zeta(s)$ is constructed.** The empirical matrix is real, non-symmetric, non-normal and dissipative, and the quantities compared with the zeros are processed arguments of its complex eigenvalues, not eigenvalues of a Hermitian operator. This paper does not construct a self-adjoint operator, a spectral determinant for $\zeta(s)$, a prime-orbit trace formula, or an argument bearing on the Riemann Hypothesis.
- **The reported fit is not uniquely reproducible.** Nominally identical unseeded optimizations of the macroscopic annealing constant k_1 converge to different local optima of a non-convex, multimodal objective (e.g., $k_1 \approx 6.48$ vs. 8.07 under Strategy C, Table 1); a reduced-scale, explicitly seeded re-run (Appendix B.5, item 4) confirms this multimodality. No single fitted value of k_1 is canonical, and the figures and tables above each explicitly state which run they use.
- **The spatial coincidence with the ion-trap data is not statistically significant.** The rank correlation between the model's residual spikes and the USTC ion-trap deviation orders is nominally significant ($\rho = 0.56$, $p = 0.010$), but the specific co-location near $N \approx 20$ is not ($p = 0.099$), and the two effects arise from distinct mechanisms (Section 4.1.5); the present model does not explain the ion-trap data.

In summary: the proposed system gives a phenomenological, calibrated reconstruction of part of the smooth counting-function behavior of the Riemann zeros, but it does not predict unseen zeros, reproduce their GUE local statistics, or furnish a self-adjoint spectral realization of the zeta function.

6.3. Future Work

From a broader perspective, this exploratory numerical study may be of interest alongside the Berry–Keating conjecture as an alternative, dissipative-dynamics numerical experiment, and it suggests a direction for exploring higher-order Riemann zeros in experimental physics. We speculate that if future quantum simulation experiments wish to reduce the current ∼1.96% dispersion envelope, it may be worth exploring whether relaxing single-sided measurement constraints and architecting full-spectrum symmetric evolution at the hardware Hamiltonian level (such as utilizing ancillary qubits to simultaneously evolve and symmetrically cancel negative energy projections) helps; we offer this as a hypothesis rather than a demonstrated prescription.

A related open question is whether the same construction could be made sensitive to zeros of Dirichlet L-functions rather than only to the Riemann zeta zeros. As presented, the cooling schedule and calibration target only the mean density of the Riemann zeros and carry no arithmetic information that would distinguish one Dirichlet character from another; the model would need some character- or conductor-dependent input (e.g., a modified map or coupling that encodes the character explicitly) before it could be expected to separate different symmetry classes. We have not attempted this and flags it as a natural, currently unaddressed extension rather than a result of the present work.

Finally, several elements of this study—the estimation of the macroscopic fitted-parameter drift with grid resolution, the macroscopic GUE ablation design, and the full-

spectrum conjugate ("vanishing fitted intercept") analysis—benefited from iterative use of large language models (LLMs) alongside our analysis, mainly for code implementation and exploratory checking. We mention this for transparency about the workflow; it does not bear on the validity of the numerical results, which stand on the released code and data.

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Data Availability Statement: All code required to reproduce the results is openly available at https://github.com/maris205/riemann_logistic (accessed on 16 August 2026). The reference Riemann zeros used for benchmarking are obtained from standard high-precision tabulations (e.g., via `mpmath.zetazero`); no restricted or proprietary data were used.

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Conflicts of Interest: The author declares no conflicts of interest.

Appendix A. Summary of the Motivating Prior Result (Ref. [31])

Because Ref. [31] is the essential motivation for the present non-autonomous map, this appendix summarizes its precise claim, the assumptions it relies on, and exactly which part of that structure is retained here.

Precise result. Ref. [31] reformulates the Sieve of Eratosthenes as a symbolic sequence over a two-letter alphabet (retained/L, eliminated/R) and, via the Metropolis–Stein–Stein (MSS) admissibility theory for unimodal maps, shows that this symbolic sequence is topologically admissible for the Logistic map $x_{n+1} = 1 - ux_n^2$ exactly at the first band-merging point $u_c \approx 1.543689$. Under this identification, the arithmetic parity constraint that consecutive integers cannot both be prime (except $2, 3$) corresponds to the forbidden word "LL" in the symbolic dynamics. To reproduce the Prime Number Theorem's asymptotic density decay ($\sim 1/\ln n$), that paper further introduces a non-autonomous "shrinking-target" mechanism, in which the effective admissible window narrows with n at a rate matched to the prime density; under this construction the Hardy–Littlewood twin-prime constant $C_2 \approx 0.6602$ emerges numerically as a fixed point of the dynamics, together with several finer statistics (discrete gap spectrum, short-range gap anti-correlation, and power-law twin-prime intermittency) that a memoryless stochastic model does not reproduce.

Assumptions. The symbolic-dynamics correspondence rests on the MSS admissibility framework for unimodal maps and on identifying the sieve's modular constraints with forbidden words at one specific, empirically located parameter value; it is a topological correspondence at that point, not a claim that holds for generic u. The shrinking-target aging schedule is fitted to match the known Prime Number Theorem density asymptotics; it is a phenomenological device for reproducing that known asymptotic, not a first-principles derivation of it.

What is retained here, and what is not. The present paper reuses only two ingredients from Ref. [31]: the same critical baseline $u_c \approx 1.5437$ as the map's operating point, and the general idea of a non-autonomous, logarithmically decaying perturbation ("aging") applied to that baseline. Everything specific to the number-theoretic result—the band-merging parity-rigidity argument, the MSS admissibility mapping to the sieve's symbol sequence, and the shrinking-target mechanism calibrated to the Prime Number Theorem—is not carried over. The cooling schedule used in the present work ($k_1/\ln^2 n + k_2/\ln^3 n$) is instead calibrated against the Riemann zero spacing (Section 2), a different target with no known relation to the prime density. The link between the two papers is, therefore, motivational: a deterministic map with a topologically distinguished critical point and a demonstrated

capacity to encode arithmetic structure at that point is a natural object to also test against the Riemann zeros, but the present numerical correspondence is not a consequence of, or extension of, the specific result proved in Ref. [31].

Appendix B. Supplementary Numerical Notes

Appendix B.1. Numerical Stability and Compiler-Induced "Butterfly Effects"

In our methodology, the non-autonomous thermodynamic evolution spans 10^{10} iterations. Due to the intrinsic nature of chaotic dynamical systems near the critical threshold (U_c), the framework exhibits extreme sensitivity to parameter precision. A macroscopic manifestation of the "butterfly effect" can be observed when switching compilation strategies during High-Performance Computing (HPC) execution.

To achieve feasible computation times across thousands of grid bins, Just-In-Time (JIT) compilation optimizations (e.g., LLVM fastmath = True) are strictly required. However, such aggressive optimizations relax the strict IEEE-754 64-bit floating-point compliance for transcendental operations (such as the logarithmic function $\ln(x)$). We observed that calculating the absolute boundary anchor (e.g., u_{temp}) entirely within the fast-math compiled kernel introduces micro-truncation errors on the magnitude of $\sim 10^{-15}$.

While negligible in standard applications, propagating this 10^{-15} discrepancy through 10^{10} non-linear mappings amplifies the error exponentially. This tiny floating-point drift in the parameter shifts the macroscopic phase space distribution, artificially inflating the Mean Squared Error (MSE) of the extracted spectral gaps by a factor of two.

Methodological Recommendation for Reproducibility

To guarantee exact spectral reconstruction and numerical stability, it is imperative to enforce a strict boundary logic separation. Foundational adiabatic parameters and baseline anchors must be pre-calculated in a strict IEEE-754-compliant environment (e.g., standard Python/NumPy interpreter). These pristine, full-precision 64-bit floating-point constants should then be explicitly passed as external arguments into the high-performance integration kernels, thereby isolating the system from compiler-induced phase drifts.

Appendix B.2. Alternative Spatial Discretization Schemes (Module A)

Before adopting the Gaussian-kernel splatter described in Section 3.2, we evaluated and iterated through three other spatial discretization schemes for mapping the infinite-dimensional continuous phase space to a finite-dimensional discrete transfer matrix:

- **Monte Carlo Trajectory:** Rationale: rather than sampling many independent points within the domain, this scheme tracks the long-range traversal behavior of a single trajectory across the time axis. Mechanism: Allowing a single particle to continuously iterate 10^7 steps in the phase space, and counting its transition frequencies between different microscopic intervals (state nodes) to construct the transfer matrix. Although intuitive, this method converges extremely slowly when handling highly localized states.
- **Naïve Point-Mapped Ulam:** Physical Intuition: The most fundamental Coarse-graining assumption. Mechanism: Dividing the phase space into equidistant grids and assuming the probability density within each grid is entirely concentrated at its geometric center. By calculating the single-mapping destination of the center point, 100% of the transition weight is assigned to the target grid. This method is simple and rough, but it introduces massive numerical truncation errors.
- **Exact Ulam's Method:** Rather than the point-particle approximation above, this method treats each grid cell as a uniform density that is stretched and folded by the

dynamical map. Mechanism: Calculating the true geometric overlap area of the source grid mapping onto the target grid through strict analytical geometry or high-precision numerical integration. The precision is extremely high, but it suffers from severe computational disasters in high dimensions or high resolutions.

The Gaussian-kernel splatter (Section 3.2) was adopted because it combines the efficiency of point mapping with the smoothing needed to suppress high-frequency spurious feature spectra caused by discretization.

Appendix B.3. Alternative Temporal Evolution Schemes (Module B)

Before adopting the time-averaged empirical matrix described in Section 2 (Module B), we considered two more direct approaches to handling the fact that the transfer matrix P_t is different at every moment in a non-autonomous system:

- **Direct Multiplication:** $\prod P_t$ rapidly leads to floating-point collapse (underflow), causing the phase space structure to instantly collapse into a trivial δ peak.
- **Continuous QR Decomposition:** Although ensuring numerical stability, its orthogonalization process discards the phase information accumulated during the system's non-autonomous evolution, since QR factors are re-orthogonalized at every step.

Both alternatives fail for the reasons given above, which is why we instead reports the spectrum of the time-averaged matrix $\bar{P}_T$ throughout, with the caveat (Section 2) that this is a heuristic surrogate for the true time-ordered-product spectrum, not a substitution justified by an adiabatic or ergodic theorem.

Appendix B.4. Full Procedure for the Microscopic ϵ-Scan

Section 4.1.2 summarizes the two-stage scan used to locate the optimal discretization scale $\epsilon \approx 0.001916$; this appendix gives the full three-step procedure. We designed a "double-layer grid scanning" experimental pipeline from macroscopic to microscopic, utilizing a 256-core parallel computing cluster to conduct an exhaustive optimization of the objective function ErrSum(ϵ).

1. Broadband Logarithmic Coarse Scan

In the first phase, to ascertain the overall parameter dependency of the system, we conducted a coarse scan with logarithmic steps within a larger parameter space $\epsilon \in [10^{-4}, 10^{-2}]$. Numerical experiments revealed the failure modes of the system at two extreme scales: when $\epsilon < 10^{-3}$ (under-smoothed limit), the rigid boundary effects of the discrete grids became prominent; the extracted eigen-phases exhibited disordered random fluctuations, and ErrSum remained persistently high, indicating that the system was overwhelmed by numerical truncation noise; when $\epsilon > 5 \times 10^{-3}$ (over-smoothed limit), the characteristic frequencies of the eigen-phases were excessively smoothed, causing the phase spacing to shrink sharply and losing the physical degrees of freedom needed to match the Riemann zeros. The coarse scan results showed that ErrSum only exhibited a distinct convergence "basin" around the 10^{-3} order of magnitude, providing a defined target zone for subsequent high-precision searching.

2. High-Density Linear Fine Scan

Based on the convergence basin located by the coarse scan, we launched a high-density linear scan with extremely small step sizes within a narrow candidate interval (e.g., $\epsilon \in [0.0018, 0.0020]$). Because a single 10^6-step evolution requires the construction of a massive time-averaged empirical transfer matrix, we mobilized multi-core computing resources for parallel computation. During this fine-tuning process, we observed a "mode-

locking" phenomenon: with minute variations in ϵ, the first six predicted energy levels progressively converge toward the true Riemann zeros.

3. Establishment of the Optimal Value

The error curve of the scanning trajectory ultimately presented a sharp "V"-shaped funnel. Numerical results indicated that when the diffusion scale advanced to $\epsilon = 0.001916$, the objective function ErrSum plummeted to its global minimum (ErrSum ≈ 2.199), shown in Figure 1. Under this smoothing scale, numerical diffusion closely offset grid truncation errors, and the dynamical system achieved close alignment with the true Riemann spectrum in the low-frequency interval ($N = 1$–6).

Appendix B.5. Additional Robustness Checks

To provide numerical evidence rather than a purely qualitative acknowledgment of open limitations, we ran four targeted follow-up experiments. We report all four here regardless of outcome; two of them surface additional, previously undocumented weaknesses rather than resolving the concerns they were designed to probe, and we do not soften that finding.

1. Sensitivity of Module C to Eigenvalue-Filtering Choices

Holding the empirical matrix fixed (single realization at $k_1 = 6.481$, 10^{10} steps, as in Strategy C of Table 1), we varied only the post-hoc spectral-extraction choices described in Appendix B.7/Module C: the magnitude cutoff ($|\lambda| \in \{0.2, 0.3, 0.4, 0.5, 0.6\}$), the conjugate-branch selection (upper half-plane, lower half-plane, or folding both halves by $|\arg \lambda|$), and whether the filtered eigenphases are sorted before unwrapping (as in the main text) or left in the native order returned by the eigensolver. The forced-origin fit reported in the main text (cutoff 0.4, upper half-plane, sorted) gives MSE ≈ 288 against the same 100 zeros used elsewhere in this comparison (a different, single-realization matrix than the one behind Table 1, so this MSE is not directly comparable to the Table 1 entries). Switching only the ordering step to "unsorted" inflates the MSE by roughly 20–40$\times$ (to $\sim$6000–11,000) and can flip the sign of the fitted slope; switching only the branch to the lower half-plane produces a similarly large degradation. The cutoff value itself matters far less over the range tested (0.2–0.5) than the sort-before-unwrap and branch choices do. We had already flagged in Appendix B.7 that the sort-then-unwrap procedure is an assumption we had not stress-tested; this experiment confirms that the procedure is not merely unverified but is in fact highly non-robust to the two discrete choices that accompany it, and the specific combination reported in the main text is doing real, load-bearing work in the fit quality.

2. Overlap Consistency Between the Gaussian-Splatting and Monte Carlo Matrix Constructions

We built both a Monte Carlo hard-count matrix (Section 4.2.4's method) and a Gaussian-kernel-splatting matrix (Module A's "Optimal Solution", Section 2) under identical physical parameters ($u_c = 1.543689$, $k_1 = 6.481$, offset 10^5) and identical, modest step/grid budgets (2×10^5 steps, 1000 bins) chosen to keep the splatting construction (which is far more expensive per step) tractable. After applying the Module C filter, the Monte Carlo matrix retained 82 eigenvalues satisfying the filter, while the Gaussian-splatting matrix retained only 3, limiting the directly shared comparison range to $N = 1$–3; the two methods' independently forced-origin-fitted slopes differed by roughly 35$\times$ (348 vs. 10) over that shared range. This experiment does not establish that the two constructions are consistent; if anything, at this step/grid scale it shows the opposite, and we cannot rule out that the

discrepancy is an artifact of the small step/grid budget imposed by the splatting method's cost. We report this as an open negative result rather than as a validated consistency check.

3. Out-of-Sample Stability of the Microscopic Discretization Scale ϵ.

Section 4.1.2's optimal $\epsilon = 0.001916$ was selected by jointly minimizing the error over all of the first six zeros, the same six zeros used to report the fit quality—a target-leakage risk. Using a reduced-cost version of the same pipeline (2×10^5 steps), we split the six zeros into a training half and a held-out half two different ways. Selecting ϵ on the odd-indexed zeros and testing on the even-indexed zeros happened to recover the same ϵ as the joint 6-zero optimum, so this split was uninformative about leakage. Selecting ϵ on the first three zeros and testing on the last three told a different story: the train-selected $\epsilon = 0.001940$ gave a held-out error sum of 28.0 on zeros 4–6, versus 5.4 when the joint ("all six zeros seen") optimum $\epsilon = 0.001804$ is evaluated on the same held-out zeros. This is direct, quantitative evidence of target leakage: an ϵ chosen without seeing the higher-order zeros generalizes to them roughly $5\times$ worse than one that was allowed to see them.

4. Seed Robustness of the k_1 Optimizer at Reduced Scale

To probe, at a computationally tractable scale, whether the differential-evolution non-reproducibility documented in Table 1's caption (Strategy C converging to $k_1 \approx 6.481$ versus $k_1 \approx 8.070$ across two unseeded runs at the full 10^{10}-step scale) reflects a genuinely multimodal objective rather than a one-off fluke, we reran the same optimizer configuration (`differential_evolution`, `strategy = 'best1bin'`, `polish = False`) at 2×10^7 steps and a reduced population (`popsize = 24`, `maxiter = 8`) under five explicit, fixed seeds. The fitted k_1 ranged from 9.33 to 14.26 across the five seeds (mean 12.43, standard deviation 1.74), confirming that the objective remains non-convex and multimodal at this reduced scale. We emphasize that these reduced-scale k_1 values are not comparable to and should not be read as confirming or replacing the full-scale 6.481/8.070 pair in Table 1: reducing the step count and grid resolution changes the objective function itself, not just its evaluation cost. What this experiment supports is the general diagnosis—an unseeded population-based optimizer on this class of objective is not expected to reproduce a single answer run-to-run—rather than a specific numerical reconciliation of the Table 1 discrepancy.

All four experiments' code and raw output are retained alongside this paper's release materials for inspection.

Appendix B.6. Dynamical Evolution Methods Approaching the Critical Point

During non-autonomous evolution, the system parameter must be annealed from the fully chaotic regime down to our empirically anchored baseline $\mu_c \approx 1.543689$ (Section 2; this is distinct from, and numerically larger than, the standard Feigenbaum accumulation point $\lambda_\infty \approx 1.401$). To investigate the decisive impact of the cooling trajectory on the resulting spectrum, we define three progressive evolution control methods:

1. Asymptotic Truncation Method

This is the conventional approach based on logarithmic cooling. The control parameter is defined as the target point plus a perturbation term decaying over time:

$$\mu(t) = U_c + \frac{k}{\ln^2(t + t_0)} \tag{A1}$$

Due to the extremely slow logarithmic decay, the parameter cannot reach U_c by 100 within a finite number of evolution steps N. In practice, this is handled by setting

an artificial microscopic truncation error (e.g., $\Delta\mu = 10^{-4}$) to deduce the constant k in combination with N. The limitation of this method is that the residual $\Delta\mu$ retains non-negligible thermal noise at the end of the system's evolution, leading to divergent high-frequency characteristics in the extracted topological spectrum.

2. Absolute Boundary Anchoring Method (1D)

To completely eliminate the truncation error, we reconstruct the control function. It does not strictly require the physical expression to include U_c directly; instead, it utilizes boundary conditions to force the system to *anchor absolutely* (100) at U_c exactly at the N-th step. The definition is as follows:

$$\mu(t) = U_{\text{temp}} + \frac{k}{\ln^2(t + t_0)} \tag{A2}$$

where the virtual base parameter U_{temp} is jointly determined by the evolution steps N and the annealing constant k: $U_{\text{temp}} = U_c - \frac{k}{\ln^2(N+t_0)}$.

Under this mechanism, defining the total steps N and a single constant k (which scales linearly with the cooling distance) guarantees that the system precisely hits the critical freezing point at the end of its long-range evolution. This allows for a close reproduction of asymptotic topological coherence within finite steps.

3. Higher-Order Anchoring Method (2D)

Although the 1D boundary anchoring resolves asymptotic tail divergence, it still encounters strong nonlinear transient fluctuations in the large-N region during early evolution (corresponding to the earliest zeros). Therefore, building upon the primary $1/\ln^2(t)$ framework, we introduce a higher-order perturbation correction term $k_2/\ln^3(t)$:

$$\mu(t) = U_{\text{temp}} + \frac{k_1}{\ln^2(t + t_0)} + \frac{k_2}{\ln^3(t + t_0)}. \tag{A3}$$

Here, U_{temp} is similarly constrained by the absolute boundary condition to ensure ultimate convergence to U_c. The introduction of k_2 is specifically designed to smooth out the curvature distortion inherent in the initial cooling phase.

Appendix B.7. Spectral Fitting and Alignment Methodology

To evaluate the degree of isomorphism between the eigenphases θ_n of the dynamical matrix and the true Riemann zero heights γ_n, this study systematically compares three fitting and evaluation criteria:

1. Conventional Linear Fitting

This approach utilizes a standard simple linear regression model:

$$\gamma_n^{\text{sim}} = a \cdot \theta_n + b \tag{A4}$$

Due to the inclusion of the intercept term b, this method possesses a higher statistical tolerance. It allows for a global translation of the spectrum, which can effectively absorb early-stage phase drifts or mask the interference of non-physical branches in the model, providing a baseline metric for the overall linear correlation.

2. Pure Scaling (Single-Degree-of-Freedom Anchoring)

This method serves as a strictly rigorous physical validation. It forcibly removes the translation degree of freedom, relying exclusively on a scaling mechanism:

$$\gamma_n^{\text{sim}} = a \cdot \theta_n \tag{A5}$$

In its strictest form, the scaling factor a is anchored solely by the first non-trivial zero ($N = 1$), such that $a = \gamma_1/\theta_1$. This pure single-point scaling requires that the intrinsic proportions of the spectral gaps within the dynamical system closely match those of the Riemann zeros: any minute distortion in these proportions will trigger exponentially growing errors as N increases. We use this strict single-point anchoring throughout, including for the $N = 1000$ dynamical fits (M1, M2) reported in Section 4.2.4, so that surviving to that scale is a genuine test of the fitted proportions rather than an artifact of a more forgiving multi-point calibration. The only exception is the GUE random-matrix baseline, which has no principled single anchor point and is instead calibrated with a global least-squares scaling $a = \sum_n \theta_n \gamma_n / \sum_n \theta_n^2$ (forced through the origin), as stated in the Figure 6 caption.

3. Piecewise Fitting via Shift-and-Invert Spectral Transformation

When scaling the alignment to extremely large datasets (e.g., $N > 10{,}000$), the computational cost and precision degradation become prohibitive. According to the Riemann–von Mangoldt formula, higher-order Riemann zeros become logarithmically denser. Attempting to extract 10,000 eigenvalues sequentially from the origin ($\theta = 0$) subjects the implicit restarted Arnoldi iteration to overwhelming low-frequency numerical noise and an $O(k^3)$ computational bottleneck, inevitably leading to system collapse.

To circumvent this curse of dimensionality, we introduce a piecewise fitting strategy utilizing the *Shift-and-Invert Spectral Transformation*. Instead of scanning the entire spectrum continuously from the lowest positive ordinate, we mathematically alter the shift parameter (σ) within the objective function. This operation dynamically transforms the transition matrix, allowing our extraction algorithm to target specific high-frequency regimes directly (e.g., the band containing the 5000th to 5100th zeros) without scanning the intervening spectrum. This technique isolates and computes chunks of high-frequency eigenstates independently. This methodology decomposes a computationally severe $O(k^3)$ problem into a series of precise, trivially parallelizable sub-tasks, ensuring numerical stability even in the deepest regimes of the energy spectrum.

Appendix B.8. Design of Ablation Studies

To rigorously validate that the precise spectral alignment originates from the intrinsic topological isomorphism of the non-autonomous dynamical map, rather than trivial mathematical coincidences, we designed a comprehensive set of ablation studies. According to the asymptotic inversion of the classical Riemann–von Mangoldt formula, the height of the n-th Riemann zero on the critical line, denoted as E_n, follows the macroscopic growth law:

$$E_n \sim \frac{2\pi n}{\ln n} \tag{A6}$$

While this fundamental equation dictates that the spacing between adjacent zeros incrementally shrinks ($\Delta E_n \sim 2\pi/\ln n$), the sequence E_n exhibits a misleading quasi-linear asymptotic trend within narrow local high-frequency bands. A naive model could theoretically yield artificially low baseline errors merely by capturing this simple linear drift over a limited interval of n. To strictly rule out this artifact, we establish a *Direct Linear Fitting* baseline, applying a standard linear regression directly to the zero index n. This

serves as the ultimate "null hypothesis," proving that our dynamical operator extracts genuine nonlinear structural information dictating the precise phase modulation, far beyond interpolating a trivial linear sequence.

Furthermore, to isolate the specific physical contributions of our proposed architecture, the model is benchmarked against two fundamental paradigms. The first is the *Static Autonomous System*, where the control parameter is fixed at the critical point ($\mu_n \equiv U_c$) without the $1/\ln^2(n)$ adiabatic cooling. This ablation specifically verifies whether the non-autonomous decay is strictly mandatory for matching the nonlinear $n/\ln n$ growth constraint. The second baseline employs classical *Random Matrix Theory (RMT)* paradigms (e.g., Gaussian Unitary Ensembles), which is known to replicate the statistical level-repulsion of the unfolded Riemann-zero spectrum, but which we do not expect to reproduce the mean counting-function trend when globally rescaled (see the discussion in Section 5.3). Contrasting the results with this GUE baseline shows that, unlike a globally rescaled random matrix, our fitted non-autonomous model tracks the mean counting-function trend numerically; we do not claim this establishes that the model captures the deterministic structure of the Riemann zeros beyond this fitted numerical trend.

Appendix B.9. Parameter Optimization Methods

To navigate the highly sensitive, non-convex parameter spaces inherent in our non-autonomous dynamical models, we employed two distinct optimization strategies tailored to different energetic regimes and computational constraints. For localized, highly sensitive critical parameters (such as the spatial discretization scale ϵ), we adopted a *Two-Stage Deterministic Scanning strategy*. This involves an initial coarse-grained broadband scan (logarithmic or linear) to rapidly identify macroscopic convergence basins, followed by a high-density, fine-grained exhaustive search within the identified optimal intervals. This "coarse-to-fine" methodology ensures that sharp, isolated topological resonances (which might easily be skipped by heuristic algorithms) are precisely locked.

For multi-dimensional or highly coupled parameter spaces (e.g., jointly fitting the macroscopic parameters k_1 and k_2), we implemented a parallelized *Differential Evolution (DE) algorithm*. Designed to handle the extreme computational overhead of large-scale evolution ($N_{\text{steps}} > 10^{10}$), the DE optimizer was configured with strict bounding priors and the best1bin mutation strategy to maximize convergence efficiency. To ensure hardware stability during massive parallel executions—specifically avoiding memory bus bottlenecks on the 128-core supercomputing cluster—the population size and parallel worker counts were strategically balanced (e.g., limiting active workers to half the physical cores). This robust heuristic approach successfully achieved global spectral alignment under stringent convergence tolerances (tol = 10^{-4}) without falling into local minima.

Appendix B.10. Reproducibility Guide: Recommended Hardware and Experimental Pipeline

All code, notebooks, and pre-computed outputs needed to reproduce every numerical claim in this paper are available at https://github.com/maris205/riemann_logistic (accessed on 16 August 2026). This subsection summarizes the recommended hardware and the main experimental steps; the repository's `readme.md` contains a complete table mapping every claim in the paper to a specific script or notebook.

Recommended hardware. No GPU is required anywhere in this study; all computations are CPU-bound integer/floating-point trajectory iteration and sparse eigendecomposition. A single modern workstation core is sufficient to reproduce every microscopic-regime result and all macroscopic-regime notebooks in minutes to tens of minutes. Two specific experiments benefit from more cores because their outer loop (over independent configurations or independent trajectory chunks) parallelizes trivially:

- The out-of-sample validation and multi-seed robustness study (Section 4.2.3) runs 13 independent configurations of 10^{10} steps each; we used a 64-core machine (`multiprocessing.Pool`, one process per configuration, `OMP_NUM_THREADS=1` inside each worker to avoid oversubscribing BLAS threads), completing in ∼176 min wall-clock. With fewer cores, wall-clock time scales roughly as $13/\min(13, n_{\text{cores}}) \times \sim 2.3$ h per configuration; every configuration is fully independent, so reducing worker count only lengthens, and never invalidates, the run.
- The large-scale ϵ-resolution scan (Section 4.1.2) and the $N = 20$ significance re-verification (Section 4.1.5) were run on 256-core and single-machine (dense $n_{\text{bins}} = 6000$ transition matrix, ∼82 min one-time build) configurations, respectively; both are embarrassingly parallel or reducible to a coarser grid if fewer cores are available, at the cost of proportionally longer wall-clock time rather than different results.

16 GB of RAM is sufficient for every experiment except the $N = 10{,}000$ ultra-macroscopic transfer matrix, which we recommend running on a machine with at least 32 GB RAM. Total disk footprint of the repository, including all pre-computed JSON/log outputs, is under 1 GB.

Software. Python 3.12 with `numpy`, `scipy`, `numba`, `mpmath`, and `matplotlib` (`pip install numpy scipy numba mpmath matplotlib`); no other dependencies are required. `numba`'s `@njit(fastmath=True)` kernels JIT-compile on first call (a few seconds of one-time overhead per script).

Main experimental steps. In the order the results appear in the paper:

1. *Microscopic regime ($N \leq 20$), optimal grid resolution ϵ* (Section 4.1.2): run the coarse scan, then the fine scan, then the figure notebook, in sequence.
2. $N = 20$ *significance test* (Section 4.1.5): run `task15_n20_significance.py`, which builds the transition matrix once and then performs 20 unselected eigendecomposition trials to quantify run-to-run variability honestly, rather than reporting a single cherry-picked trial.
3. $N = 100$ *anchoring-strategy ablation* (Section 4.2.2): run the four Model A–D notebooks independently, then run the overlay notebook that reproduces Figure 4.
4. *Out-of-sample validation and multi-seed robustness* (Section 4.2.3): run `task12_13_experiments_v2.py`, which reproduces all 13 JSON result files (3 held-out splits $\times$ evaluation, plus 10 seeds at fixed $M = 70$).
5. $N = 1000$ *macroscopic fit* (Section 4.2.4, Figure 6): run the corresponding notebook, which performs the strict single-point anchoring described in Appendix B.7 for models M1/M2 and the forced-origin global scaling for the GUE baseline.
6. *Unfolded local-statistics GUE comparison* (Section 5.3): run `task14_rmt_unfolded.py` (true Riemann zeros vs. GUE/Poisson) and `task14_dynamical_rmt.py` (the dynamical model's own eigenphases vs. GUE/Poisson).
7. *Additional robustness checks* (Appendix B.5): run, from the `review_experiments/` subdirectory, `task24_phase_extraction_sensitivity.py`, `task25_splatting_vs_mc_overlap.py`, `task26_epsilon_out_of_sample.py`, and `task27_de_seed_robustness_reduced.py` independently; each writes its own JSON results file and none depends on the others' output. These four scripts were run at deliberately reduced scale relative to the rest of the pipeline (see each script's header for the exact step/grid/population reduction and the rationale) to keep the turnaround tractable on a single 64-core workstation, so their absolute numbers are not directly comparable to the full-scale results elsewhere in the paper; only their qualitative conclusions are used in the main text.

Every `task*.py` script locates its own output directory relative to its own file path, so it can be re-run from a fresh clone without editing any paths. Pre-computed outputs

are committed alongside each script, so every numerical claim can be inspected without re-running anything, or verified by re-running and comparing.

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